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(12) **United States Patent**
Hasenberg et al.

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(54) **SYSTEMS, DEVICES, AND METHODS FOR GENERATING SHOCK WAVES IN A FORWARD DIRECTION**

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(73) Assignee: **SHOCKWAVE MEDICAL, INC.**, Santa Clara, CA (US)

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A61M 5/00 (2006.01)
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(52) **U.S. Cl.**
CPC ***A61B 17/22012*** (2013.01); *A61B 2017/00137* (2013.01); *A61B 2017/00292* (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC A61B 17/22012; A61B 2017/00137; A61B 2017/00292; A61B 2017/00982; A61B 2017/22021; A61B 2218/007
See application file for complete search history.

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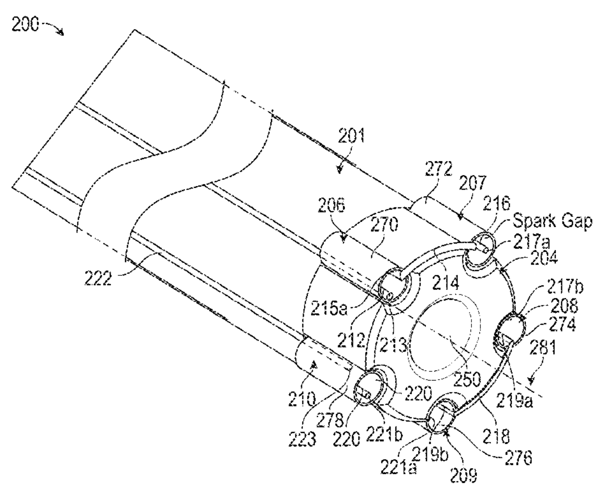
(74) *Attorney, Agent, or Firm* — Morrison & Foerster LLP

(57) **ABSTRACT**

An exemplary catheter for use in a body lumen comprises: a catheter body; and a plurality of shock wave emitters disposed at a distal end of the catheter body, each shock wave emitter configured to generate a shock wave that propagates distally of the catheter body, wherein the plurality of shock wave emitters are arrayed about a longitudinal axis of the catheter body such that shock waves emitted from the plurality of shock wave emitters can constructively interfere distally of the catheter body, and wherein each shock wave emitter comprises electrodes separated by a spark gap and at least one electrical connector that connects at least one of the electrodes to an electrode of another shock wave emitter of the plurality of shock wave emitters.

61 Claims, 15 Drawing Sheets



(51) **Int. Cl.**
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(52) **U.S. Cl.**
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Invitation to pay additional fees received for International Patent Application No. PCT/US2023/082079 mailed on Aug. 12, 2024, 3 pages.

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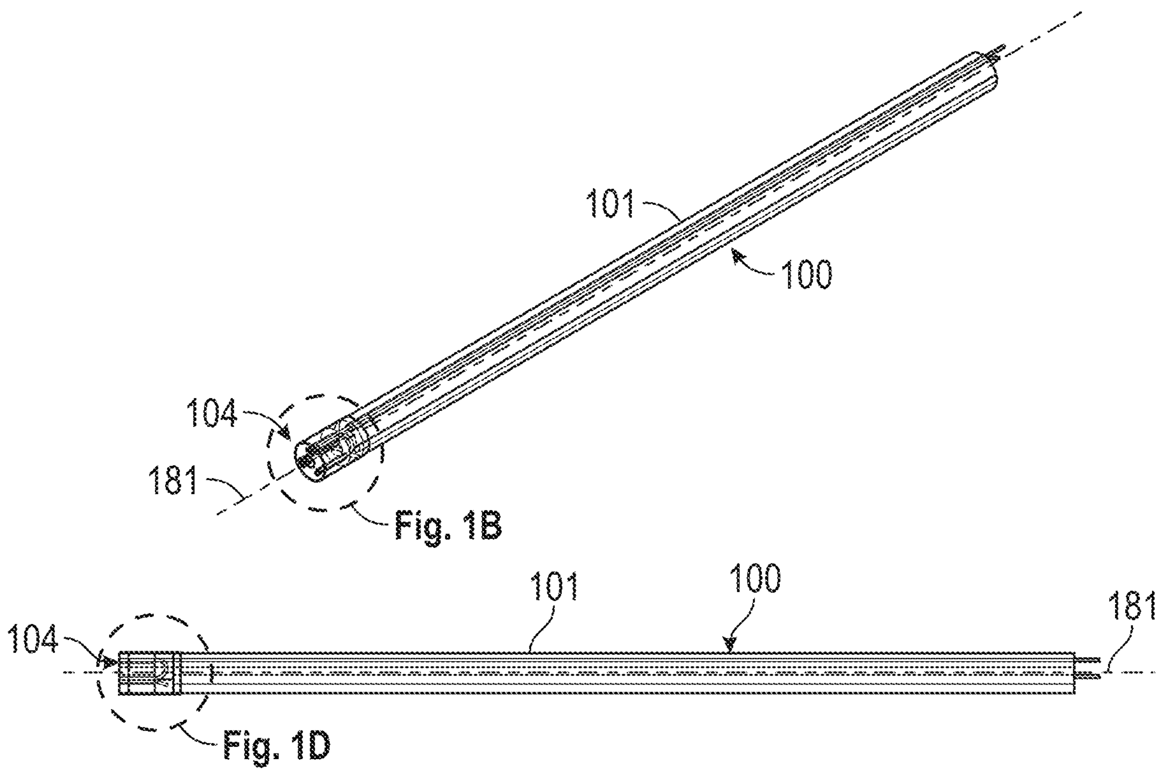
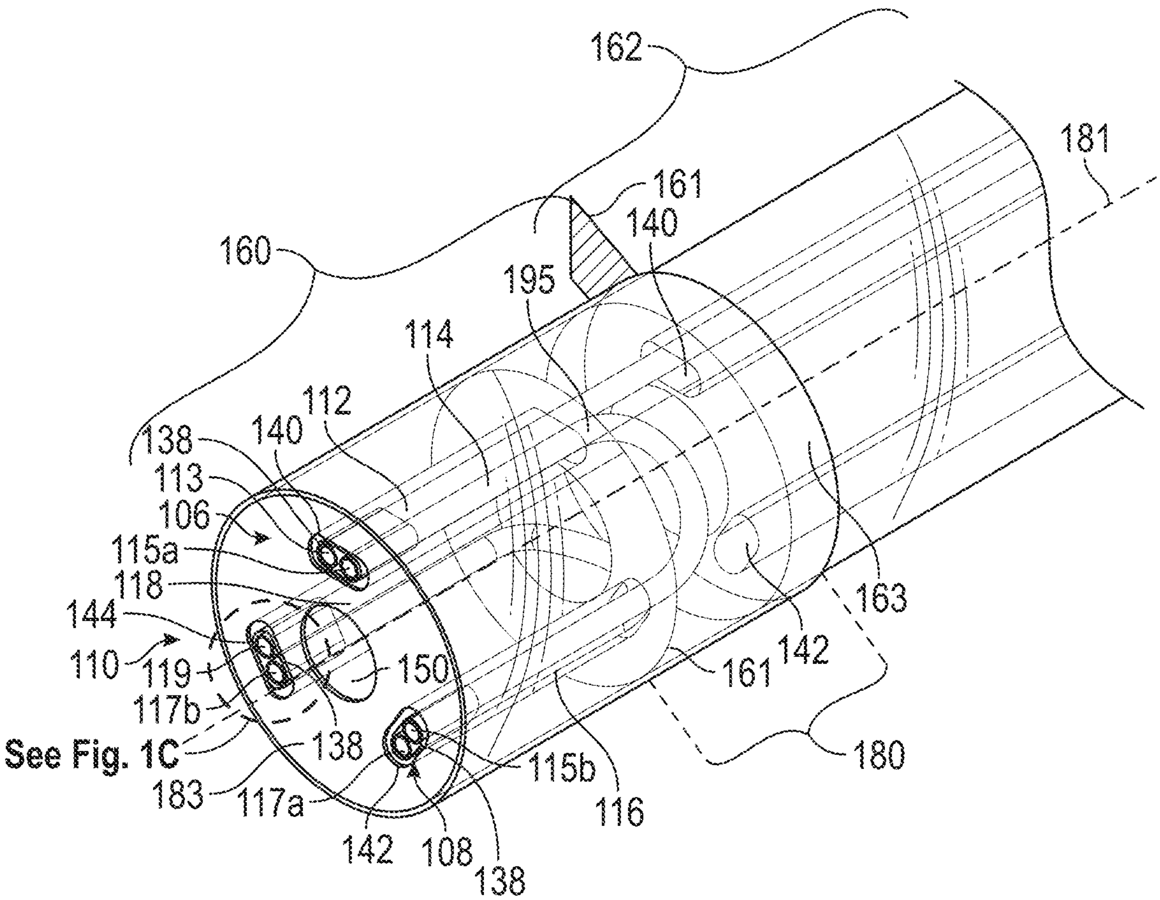


FIG. 1A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113
106
115a
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144
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119
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117b
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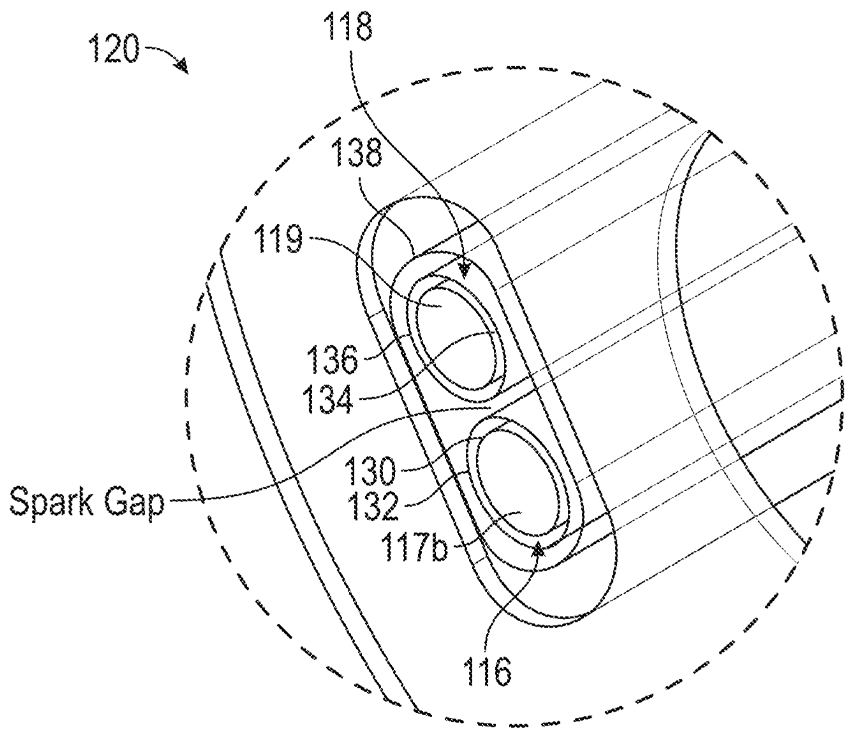
183
117a
142
108
138
115b
116
161
142
163
180

FIG. 1B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Spark Gap
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132
117b
116

FIG. 1C

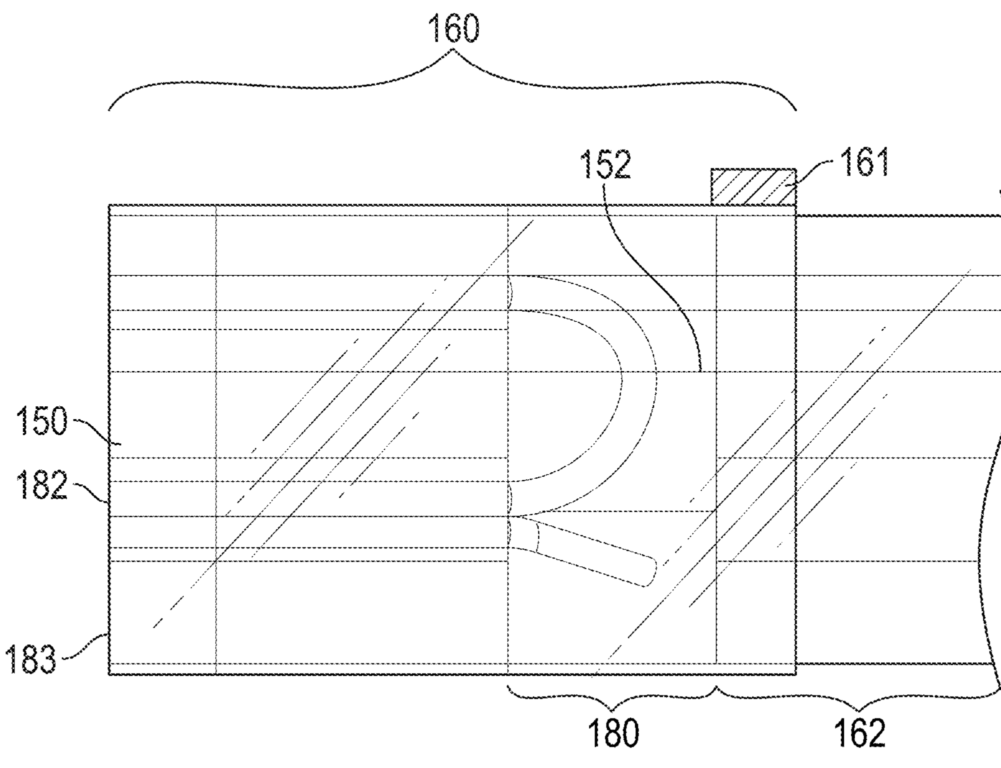
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FIG. 1D

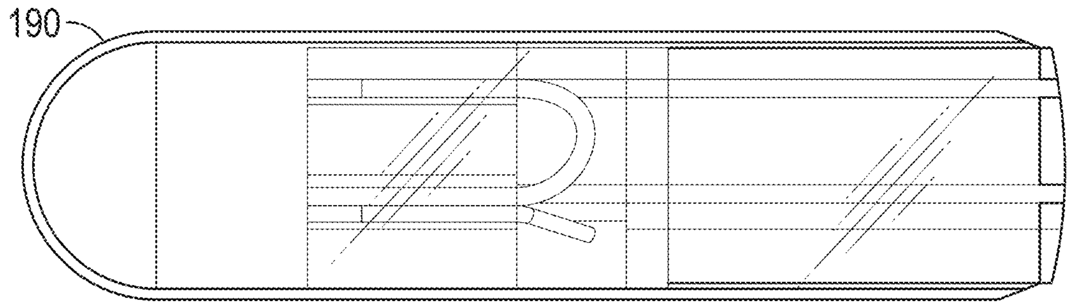


FIG. 2A

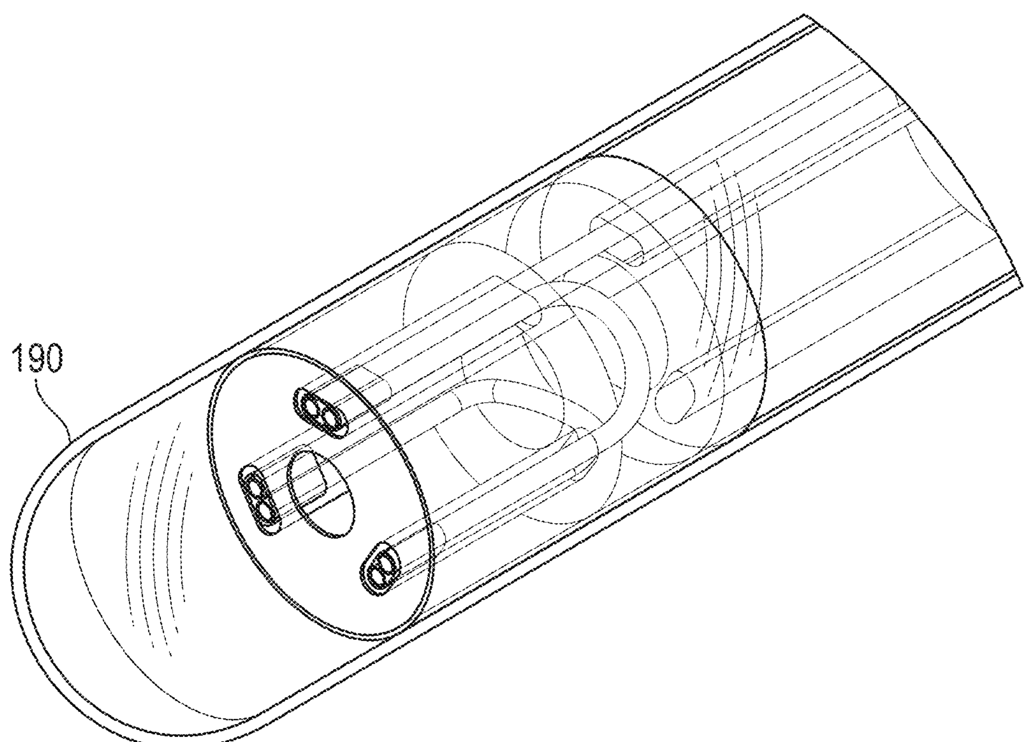
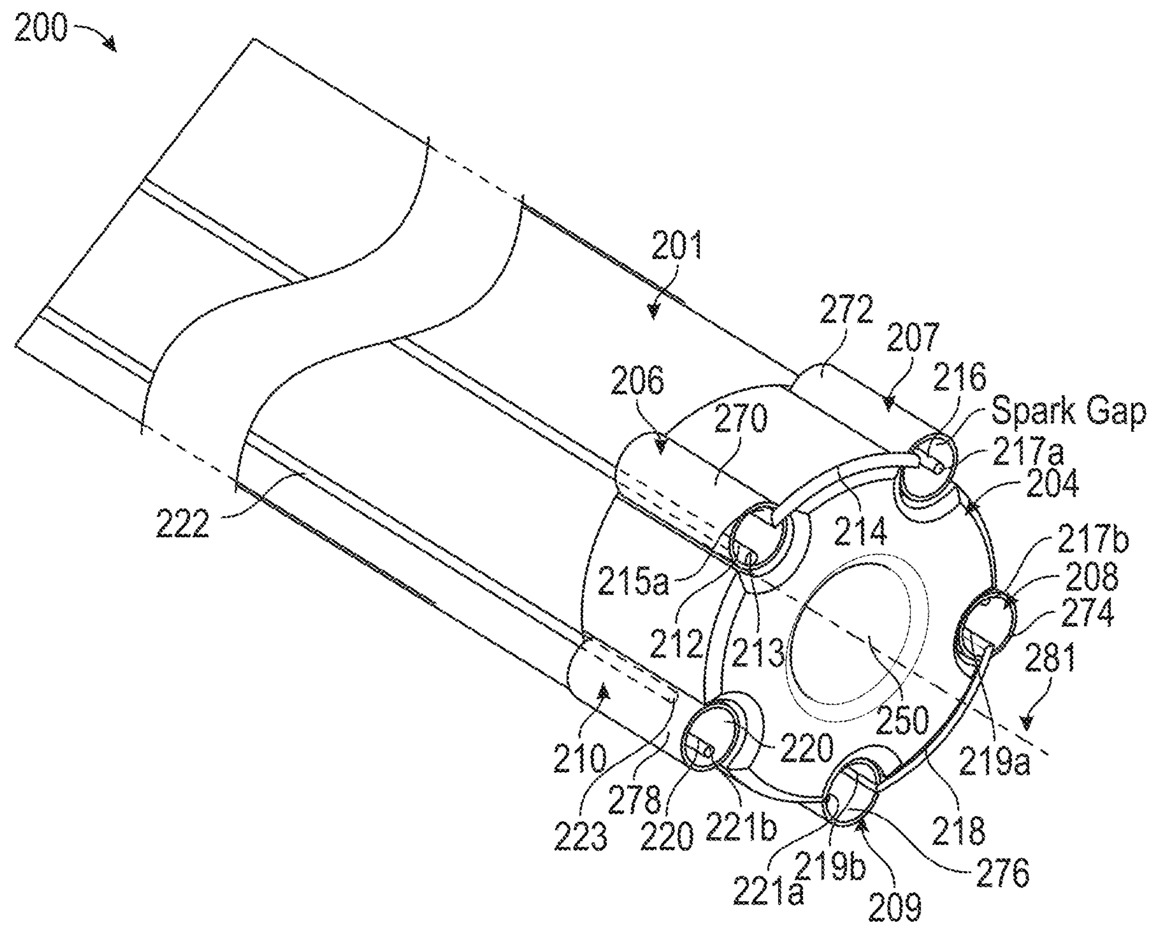


FIG. 2B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216
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222
217b
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215a
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281
250
210
220
219a
278
218
223
220
221b
219b
276
221a
209

FIG. 3A

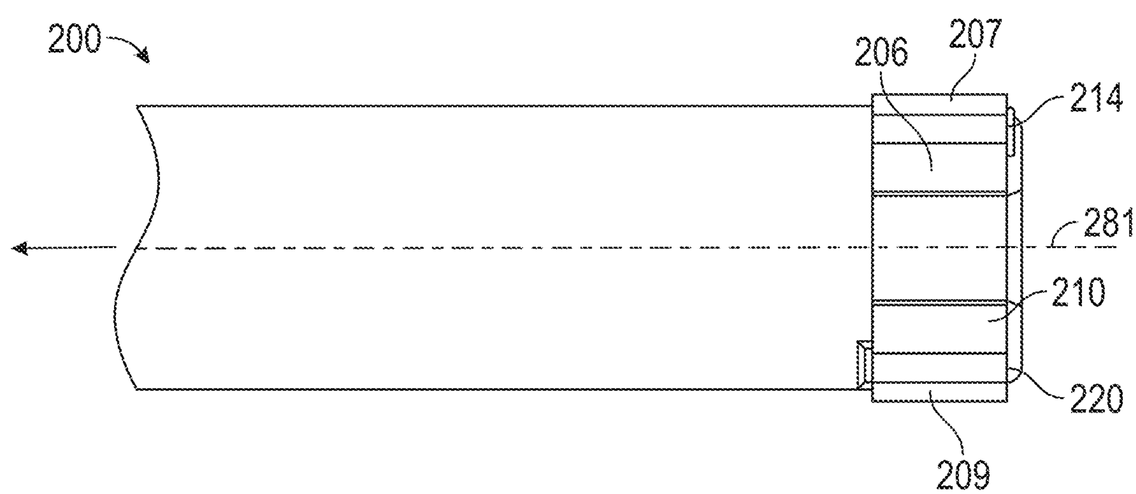
200
207
206
214
281
210
220
209

FIG. 3B

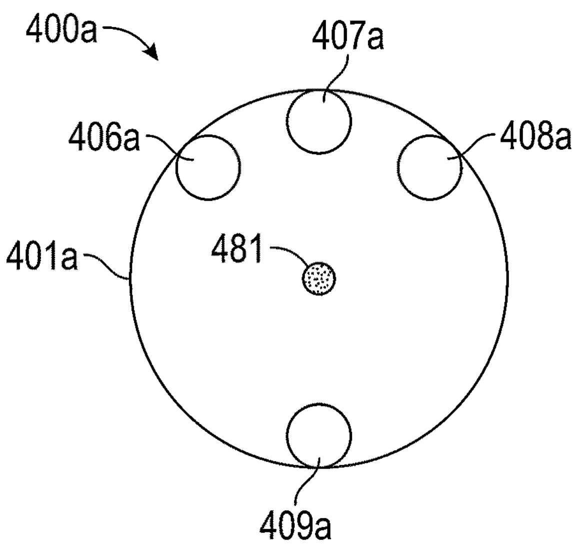

FIG. 4A

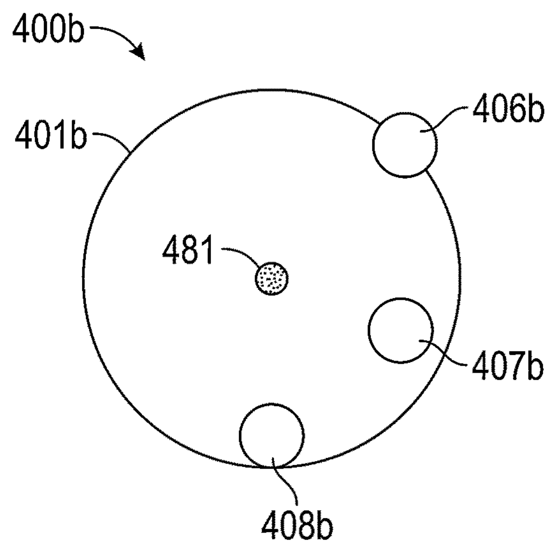

FIG. 4B

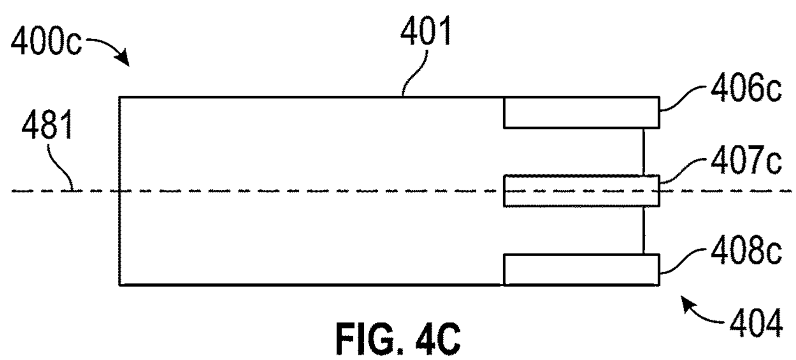

FIG. 4C

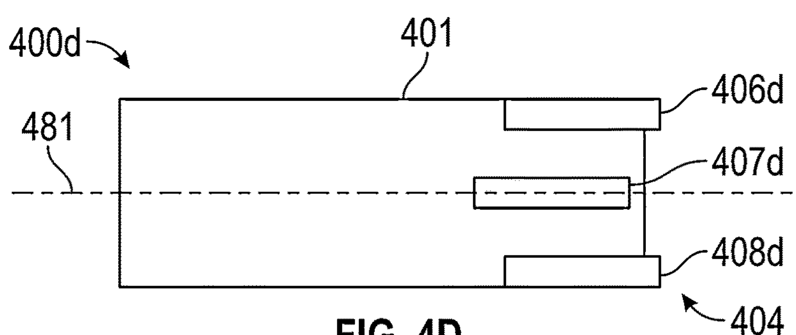

FIG. 4D

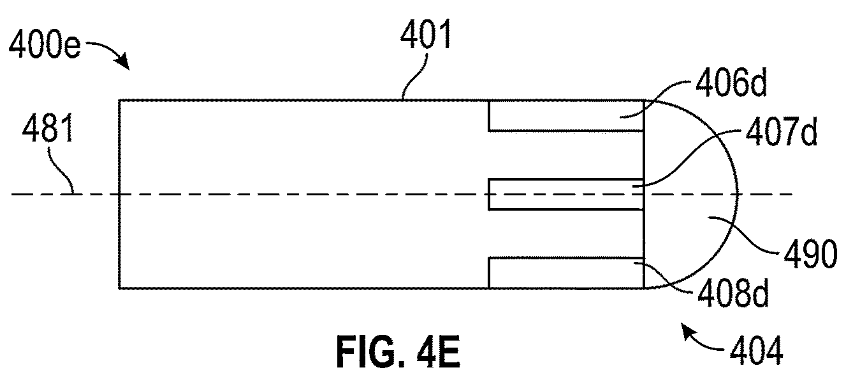

FIG. 4E

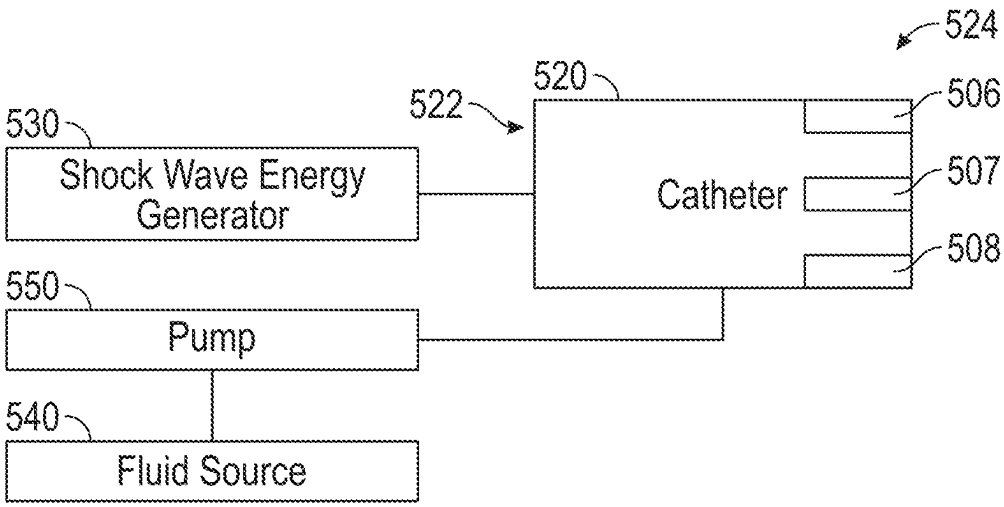
524
522
520
506
507
508
530
Shock Wave Energy Generator
Catheter
550
Pump
540
Fluid Source

FIG. 5

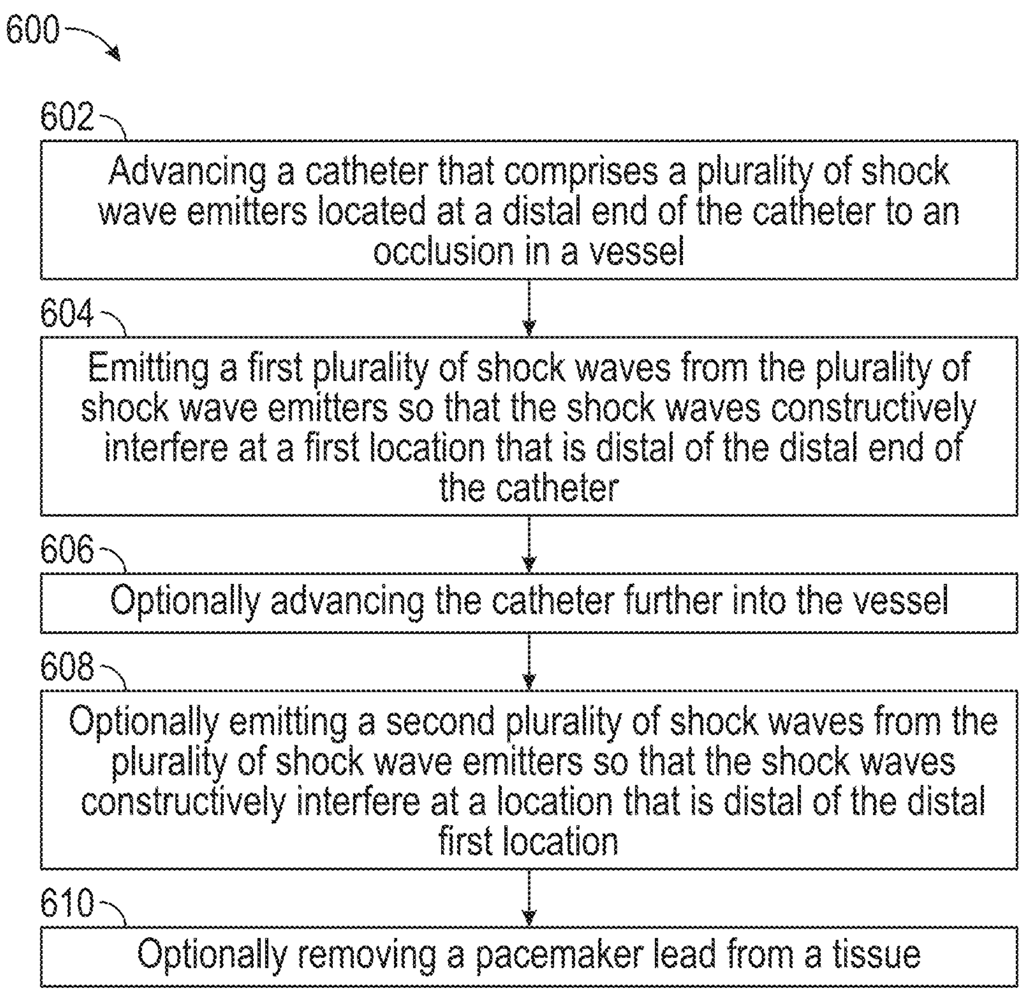
600
602
Advancing a catheter that comprises a plurality of shock wave emitters located at a distal end of the catheter to an occlusion in a vessel
604
Emitting a first plurality of shock waves from the plurality of shock wave emitters so that the shock waves constructively interfere at a first location that is distal of the distal end of the catheter
606
Optionally advancing the catheter further into the vessel
608
Optionally emitting a second plurality of shock waves from the plurality of shock wave emitters so that the shock waves constructively interfere at a location that is distal of the distal first location
610
Optionally removing a pacemaker lead from a tissue

FIG. 6

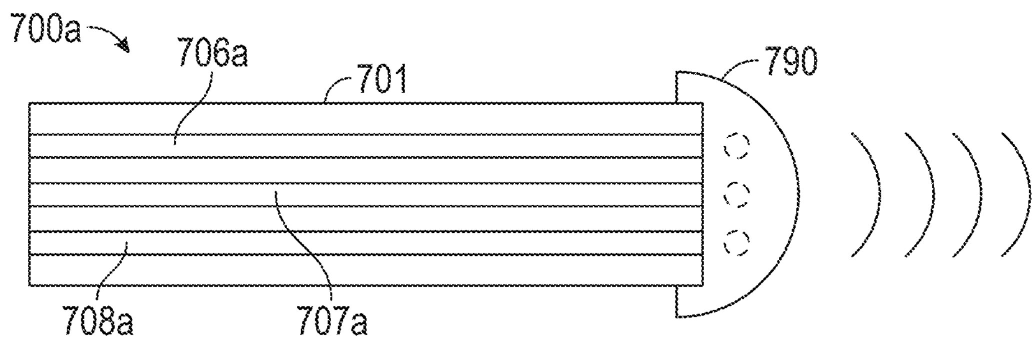

FIG. 7A

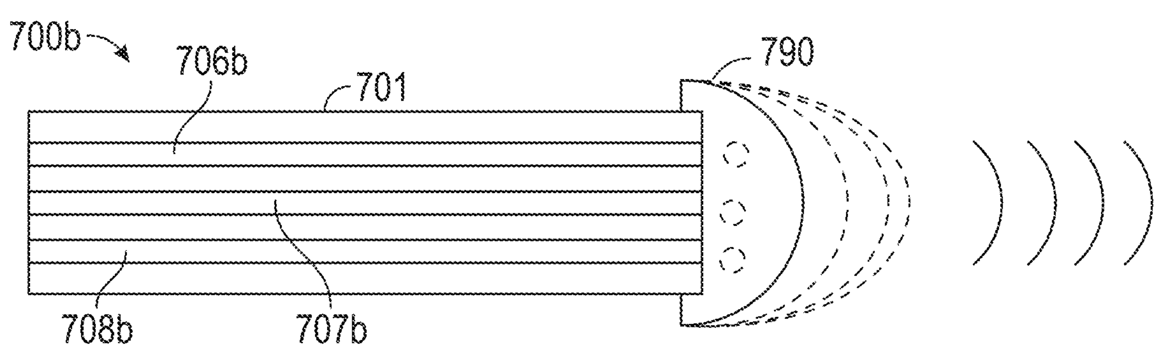

FIG. 7B

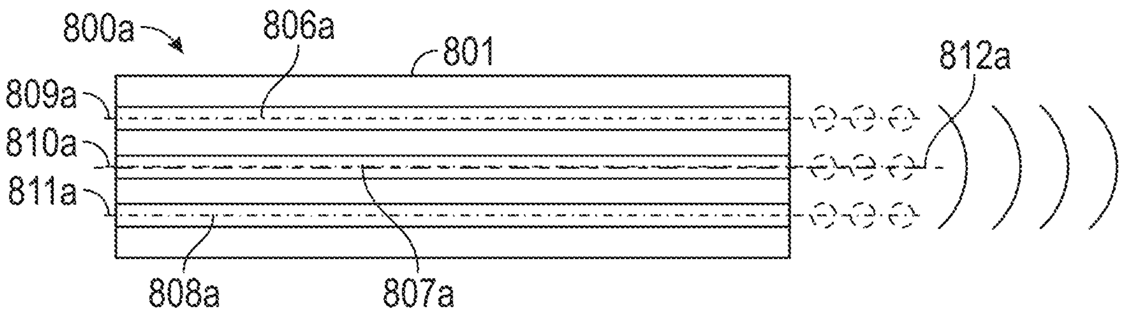

FIG. 8A

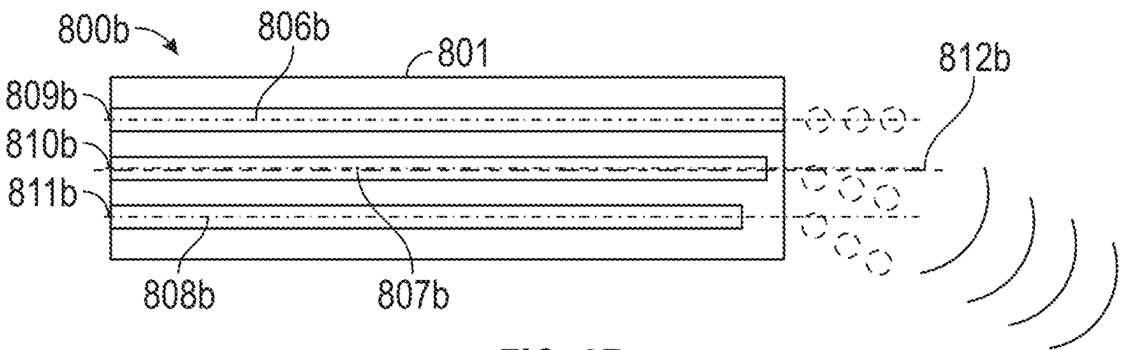

FIG. 8B

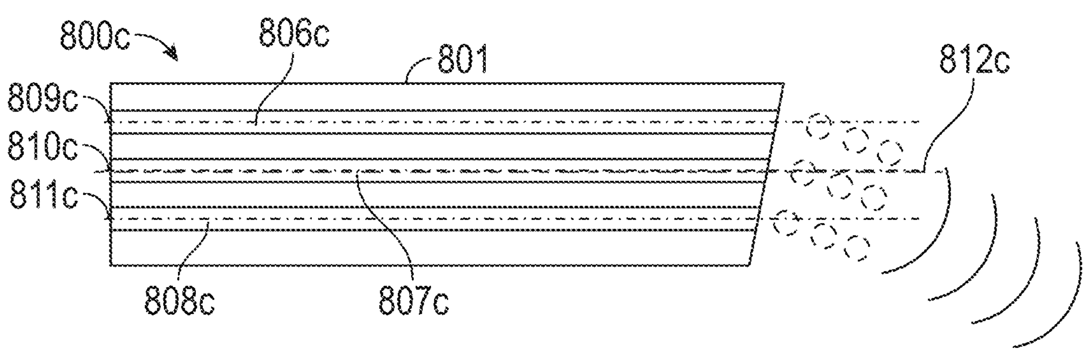

FIG. 8C

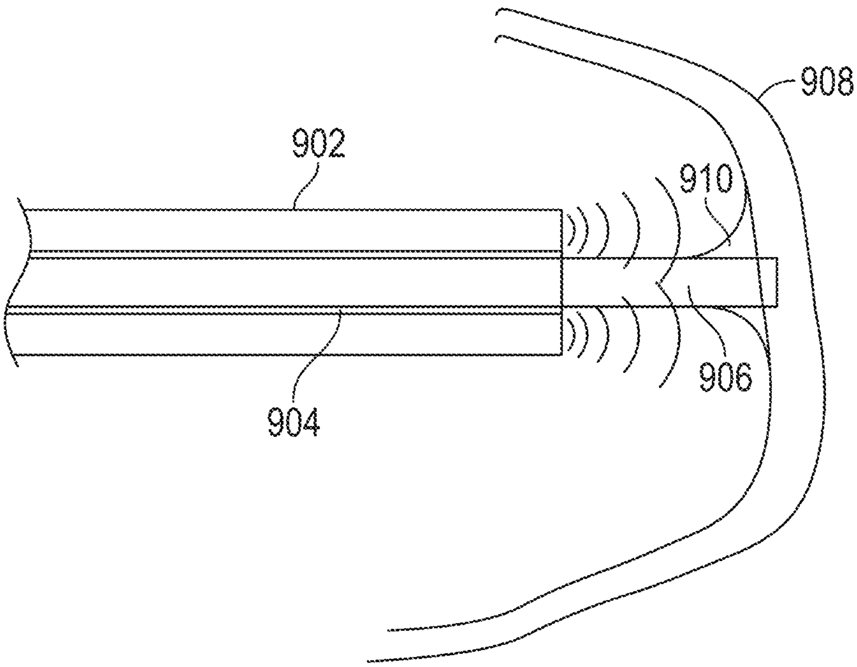
908
902
910
906
904

FIG. 9

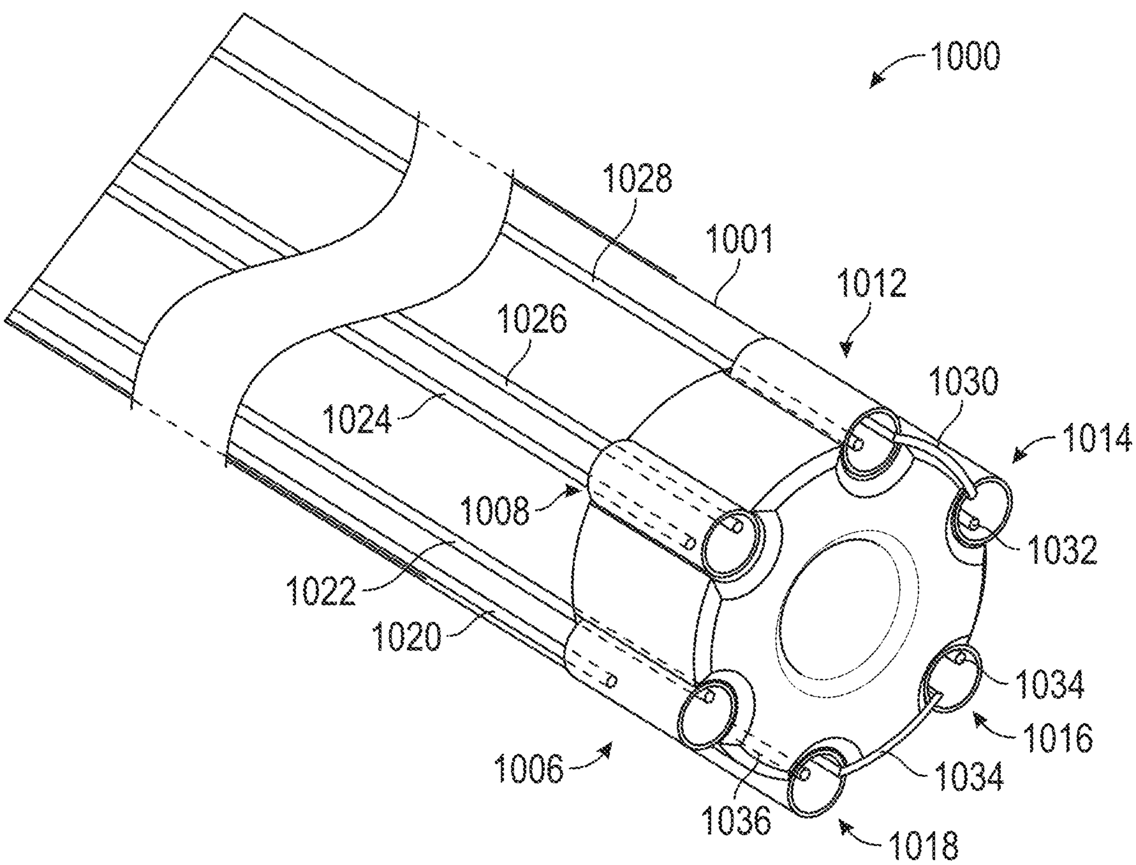
1000
1028
1001
1012
1026
1030
1014
1024
1008
1032
1022
1020
1034
1016
1006
1034
1036
1018

FIG. 10

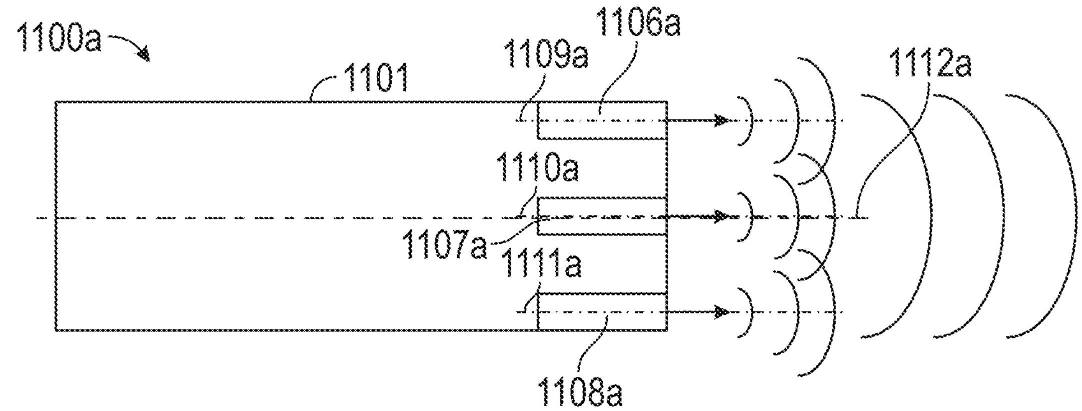

FIG. 11A

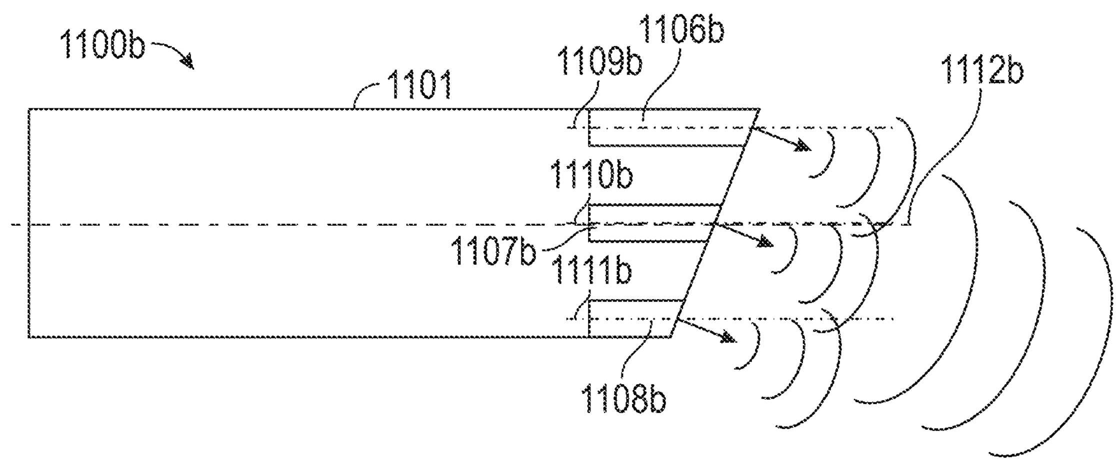

FIG. 11B

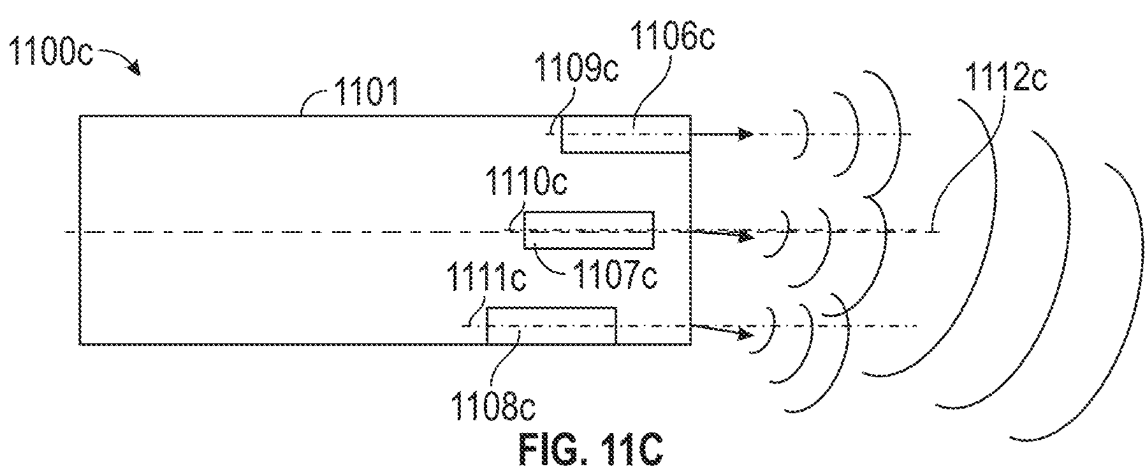

FIG. 11C

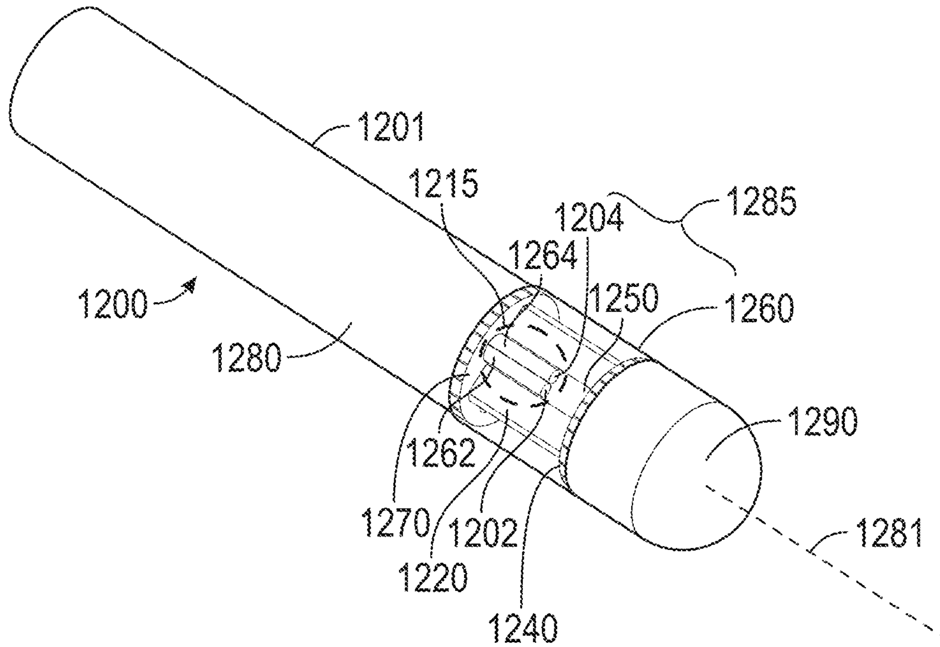

FIG. 12A

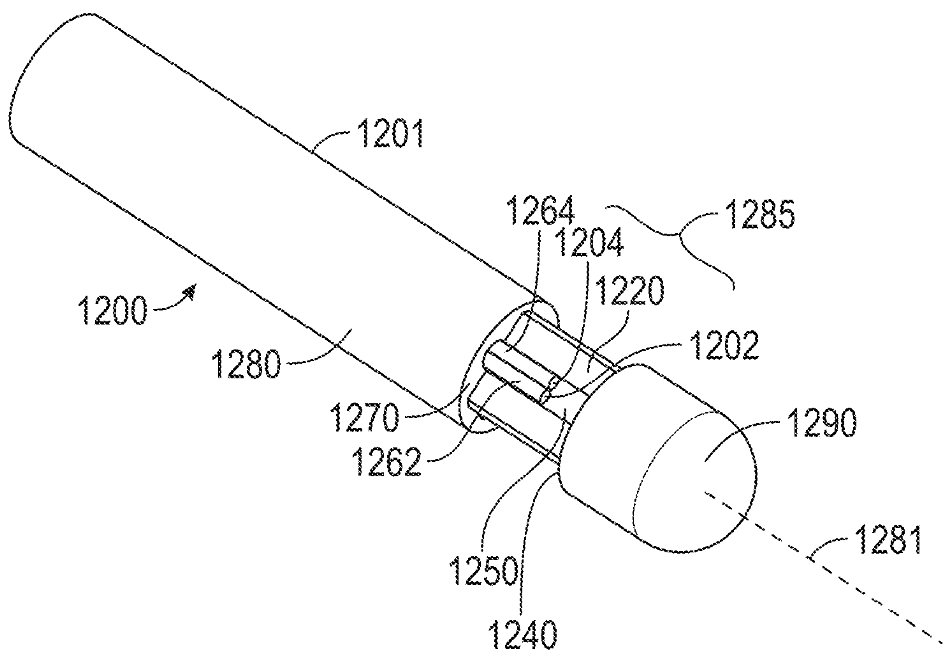

FIG. 12B

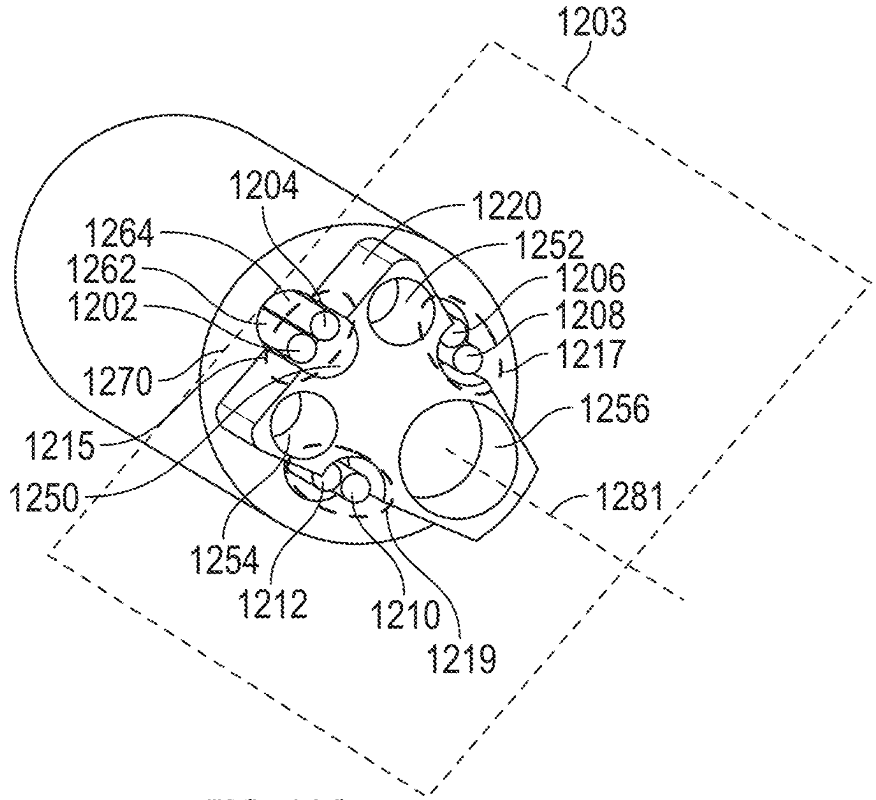

FIG. 12C

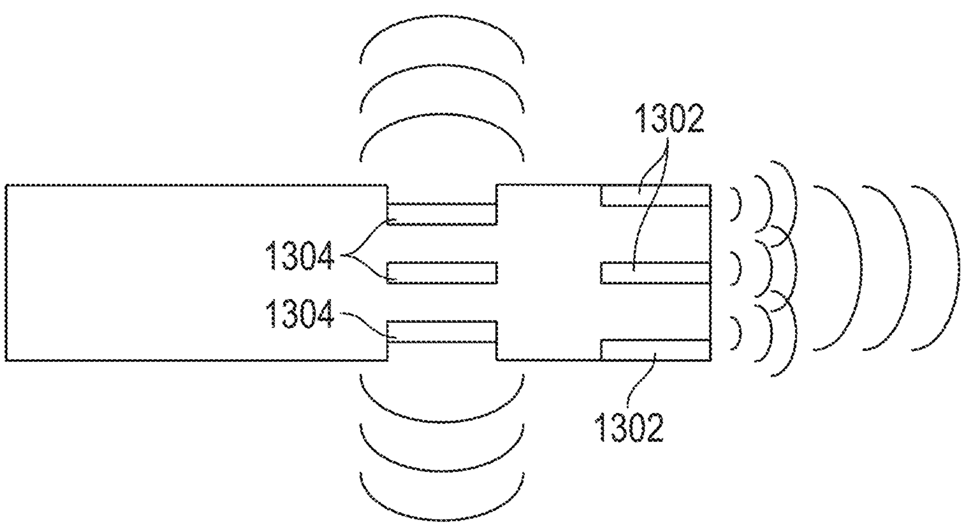

FIG. 13A

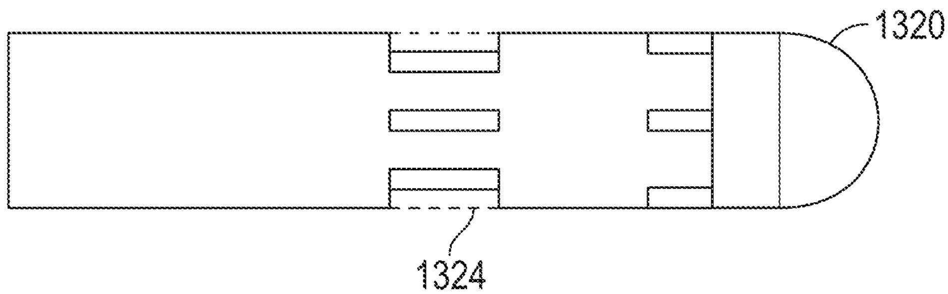

FIG. 13B

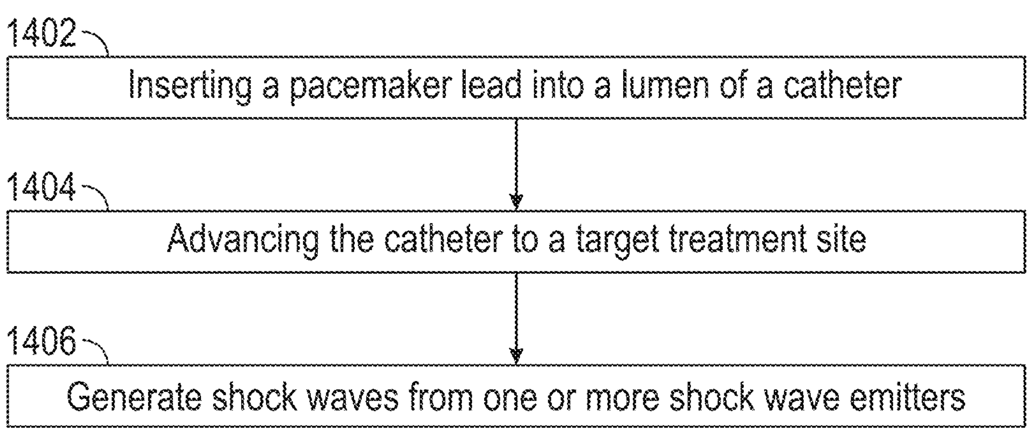

FIG. 14

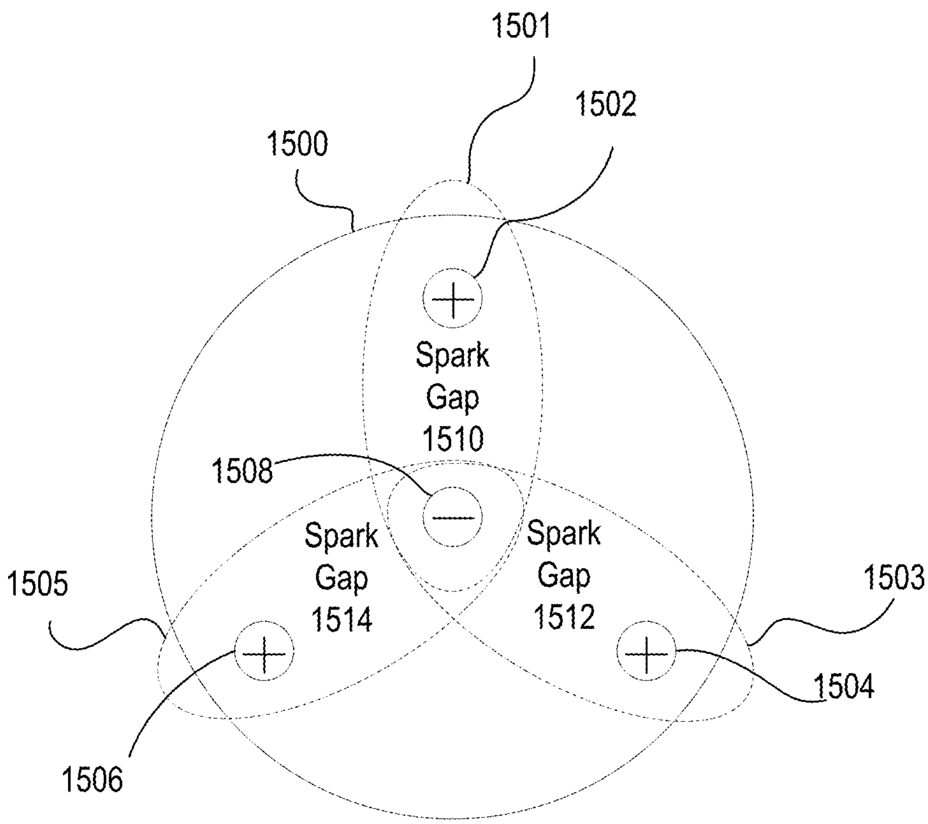
1501
1502
1500
Spark
Gap
1510
1508
Spark
Gap
1514
Spark
Gap
1512
1505
1503
1504
1506

FIG. 15

SYSTEMS, DEVICES, AND METHODS FOR GENERATING SHOCK WAVES IN A FORWARD DIRECTION

FIELD

The present disclosure relates generally to the field of medical devices and methods, and more specifically to shock wave catheter devices for treating calcified lesions in body lumens, such as calcified lesions and occlusions in vasculature and kidney stones in the urinary system.

BACKGROUND

A wide variety of catheters have been developed for treating calcified lesions, such as calcified lesions in vasculature associated with arterial disease. For example, treatment systems for percutaneous coronary angioplasty or peripheral angioplasty use angioplasty balloons to dilate a calcified lesion and restore normal blood flow in a vessel. In these types of procedures, a catheter carrying a balloon is advanced into the vasculature along a guide wire until the balloon is aligned with calcified plaques. The balloon is then pressurized (normally to greater than 10 atm), causing the balloon to expand in a vessel to push calcified plaques back into the vessel wall and dilate occluded regions of vasculature.

More recently, the technique and treatment of intravascular lithotripsy (IVL) has been developed, which is an interventional procedure to modify calcified plaque in diseased arteries. The mechanism of plaque modification is through use of a catheter having one or more acoustic shock wave generating sources located within a liquid that can generate acoustic shock waves that modify the calcified plaque. IVL devices vary in design with respect to the energy source used to generate the acoustic shock waves, with two exemplary energy sources being electrohydraulic generation and laser generation.

For electrohydraulic generation of acoustic shock waves, a conductive solution (e.g., saline) may be contained within an enclosure that surrounds electrodes or can be flushed through a tube that surrounds the electrodes. The calcified plaque modification is achieved by creating acoustic shock waves within the catheter by an electrical discharge across the electrodes. The energy from this electrical discharge enters the surrounding fluid faster than the speed of sound, generating an acoustic shock wave. In addition, the energy creates one or more rapidly expanding and collapsing vapor bubbles that generate secondary shock waves. The shock waves propagate radially outward and modify calcified plaque within the blood vessels. For laser generation of acoustic shock waves, a laser pulse is transmitted into and absorbed by a fluid within the catheter. This absorption process rapidly heats and vaporizes the fluid, thereby generating the rapidly expanding and collapsing vapor bubble, as well as the acoustic shock waves that propagate outward and modify the calcified plaque. The acoustic shock wave intensity is higher if a fluid is chosen that exhibits strong absorption at the laser wavelength that is employed. These examples of IVL devices are not intended to be a comprehensive list of potential energy sources to create IVL shock waves.

The IVL process may be considered different from standard atherectomy procedures in that it cracks calcium but does not liberate the cracked calcium from the tissue. Hence, generally speaking, IVL should not require aspiration nor embolic protection. Further, due to the compliance of a normal blood vessel and non-calcified plaque, the shock waves produced by IVL do not modify the normal vessel tissue or non-calcified plaque. Moreover, IVL does not carry the same degree of risk of perforation, dissection, or other damage to vasculature as atherectomy procedures or angioplasty procedures using cutting or scoring balloons.

More specifically, catheters to deliver IVL therapy have been developed that include pairs of electrodes for electrohydraulically generating shock waves inside an angioplasty balloon. Shock wave devices can be particularly effective for treating calcified plaque lesions because the acoustic pressure from the shock waves can crack and disrupt lesions near the angioplasty balloon without harming the surrounding tissue. In these devices, the catheter is advanced over a guidewire through a patient's vasculature until it is positioned proximal to and/or aligned with a calcified plaque lesion in a body lumen. The balloon is then inflated with conductive fluid (using a relatively low pressure of 2-4 atm) so that the balloon expands to contact the lesion but is not an inflation pressure that substantively displaces the lesion. Voltage pulses can then be applied across the electrodes of the electrode pairs to produce acoustic shock waves that propagate through the walls of the angioplasty balloon and into the lesions. Once the lesions have been cracked by the acoustic shock waves, the balloon can be expanded further to increase the cross-sectional area of the lumen and improve blood flow through the lumen. Alternative devices to deliver IVL therapy can be within a closed volume other than an angioplasty balloon, such as a cap, balloons of variable compliancy, or other enclosure.

Calcium buildup in various body structures, such as Mitral Annular Calcification (MAC) and Chronic Total Occlusions (CTOs), can become very thick and thus difficult or even impossible to treat using known methods. As described above, a relatively novel approach for treating such calcification includes introducing a catheter into the affected structure and generating shock waves within the body structure using the catheter to break up the calcifications. Existing shock wave devices typically include shock wave emitters spaced along the length (i.e., along the longitudinal axis) of the device's body and are used to treat buildup of calcified plaque along the length of the inner wall of a body lumen such as a blood vessel. Such devices are not configured for generating shock waves in a forward direction. Shock wave devices configured for forward firing exist, but are not capable of withstanding high voltages, for instance up to 20 kV, and are thus limited in their ability to generate powerful shock waves in the forward direction. Further, known devices include a limited number of forward shock wave emitters (e.g., one or two), which minimizes the constructive interference of the shock waves distally of the end of the device, thus further reducing the capacity of the device to break up dense calcifications.

SUMMARY

Described herein are systems, devices, and methods for generating shock waves that propagate in a substantially forward direction from a distal end of a catheter, for instance, to treat calcified plaques or other calcifications and obstructions in the body that are distal of the distal end of the catheter. In some embodiments, the catheter includes a plurality of shock wave emitters arrayed about a longitudinal axis at the distal end of the catheter. Each of the shock wave emitters includes an electrode pair separated by a spark gap for generating shock waves that propagate forward of the catheter. The forward directed shock waves generated by the shock wave emitters can constructively interfere with one another distally of the distal end of the catheter to produce a powerful peak compressive force for treating dense calcifications, such as Mitral Annular Calcification (MAC) and Chronic Total Occlusions (CTOs).

In some embodiments, the catheters described herein can be inserted into a body lumen such as a blood vessel until the distal end of the catheter is positioned such that a target treatment area in the body lumen is positioned at least partially forward of the distal end of the catheter. Voltage pulses can be applied to a plurality of shock wave emitters located at the distal end of the catheter to generate shock waves directed forward toward the target treatment area. In some embodiments, the shock wave emitters may be encased in an enclosure, such as a cap or balloon that can be filled with a conductive fluid. In some embodiments, the shock wave emitters are not enclosed, and vapor bubbles (e.g., cavitation bubbles) generated by energy discharged from the shock wave emitters can contribute to the treatment of the target area.

According to some aspects, a catheter for use in a body lumen includes a catheter body; and a plurality of shock wave emitters disposed at a distal end of the catheter body, each shock wave emitter configured to generate a shock wave that propagates distally of the catheter body, and wherein at least one shock wave emitter of the plurality of shock wave emitters comprises electrodes separated by a spark gap and at least one electrical connection to an electrode of at least one other shock wave emitter of the plurality of shock wave emitters.

Optionally, the plurality of shock wave emitters are arranged such that shock waves emitted from the plurality of shock wave emitters can constructively interfere distally of the catheter body. Optionally, the plurality of shock wave emitters are electrically connected in series such that an electrical pulse applied across an electrode of a first shock wave emitter of the plurality of shock wave emitters and an electrode at a second shock wave emitter of the plurality of shock wave emitters causes each of the plurality of shock wave emitters to emit a respective shock wave.

Optionally, at least one shock wave emitter of the plurality of shock wave emitters can be driven independently of at least a second shock wave emitter of the plurality of shock wave emitters. Optionally, a first shock wave emitter of the plurality of shock wave emitters comprises an exposed distal tip of a first insulated wire and an exposed distal tip of a second insulated wire, the second insulated wire extending to a second shock wave emitter of the plurality of shock wave emitters.

Optionally, the first insulated wire extends along the length of the catheter body and is positioned within a lumen of the catheter body. Optionally, a first shock wave emitter of the plurality of shock wave emitters comprises an exposed distal end of an insulated wire and a conductive emitter band that is separated from the exposed distal end of the insulated wire by a spark gap.

Optionally, the plurality of shock wave emitters comprises at least three shock wave emitters. Optionally, the plurality of shock wave emitters are arrayed symmetrically about a longitudinal axis of the catheter body. Optionally, a distal most surface of a shock wave emitter of the plurality of shock wave emitters is flush with a distal most surface of the distal end of the catheter body.

Optionally, a distal most surface of a shock wave emitter of the plurality of shock wave emitters is recessed from a distal most surface of the distal end of the catheter body. Optionally, a distal most surface of a shock wave emitter of the plurality of shock wave emitters is positioned forward of a distal most surface of the distal end of the catheter body.

Optionally, the plurality of shock wave emitters are disposed at the same distal location relative to the distal end of the catheter body. Optionally, an outermost surface of a shock wave emitter of the plurality of shock wave emitters is inset relative to an outer circumferential surface of the catheter body. Optionally, an outermost surface of a shock wave emitter of the plurality of shock wave emitters is positioned externally relative an outer circumferential surface of the catheter body.

Optionally, the catheter includes an enclosure positioned to cover the plurality of shock wave emitters at the distal end of the catheter body. Optionally, the enclosure is configured to be filled with a conductive fluid. Optionally, the catheter includes a central lumen extending from the proximal end of the catheter to the distal end of the catheter.

Optionally, the central lumen is configured to receive a guide wire. Optionally, the central lumen is configured to receive a pacemaker wire lead. Optionally, the catheter body comprises an aspiration lumen. Optionally, the plurality of shock wave emitters are respectively spaced apart from one another by a distance of between 1 mm and 10 mm.

According to some aspects, a catheter for use in a body lumen comprises: a catheter body; and at least three shock wave emitters disposed at a distal end of the catheter body, each shock wave emitter configured to generate a shock wave that propagates distally of the catheter body, wherein the at least three shock wave emitters are arrayed about a longitudinal axis of the catheter body such that shock waves emitted from the at least three shock wave emitters can constructively interfere distally of the catheter body, wherein each shock wave emitter comprises electrodes separated by a spark gap. Optionally, the at least three shock wave emitters are all electrically connected in series such that an electrical pulse applied to a first shock wave emitter of the at least three shock wave emitters causes each of the at least three shock wave emitters to emit a respective shock wave. Optionally, at least a first shock wave emitter of the at least three shock wave emitters can be driven independently of at least a second shock wave emitter of the at least three shock wave emitters. Optionally, each shock wave emitter of the at least three shock wave emitters shares a common electrode with each of the other shock wave emitters. Optionally, the common electrode is positioned at an equal distance from an electrode of each of the at least three shock wave emitters.

According to some aspects, a shock wave generating system comprises: a shock wave energy generator; and a catheter comprising: a catheter body; and a plurality of shock wave emitters disposed at a distal end of the catheter body, each shock wave emitter configured to generate a shock wave that propagates distally of the catheter body, wherein the plurality of shock wave sources are arranged such that shock waves emitted from the plurality of shock wave emitters can constructively interfere distally of the catheter body, and wherein each shock wave emitter comprises electrodes separated by a spark gap and at least one electrical connector that connects at least one of the electrodes to an electrode of another shock wave emitter of the plurality of shock wave emitters. Optionally, the shock wave energy generator is configured to deliver high voltage pulses to a shock wave emitter of the plurality of shock wave emitters, wherein the high voltage pulses are between 3 kV and 20 kV, including 3 kV and 20 kV. Optionally, the shock wave energy generator is configured to deliver the voltage pulses at a rate of up to 20 Hz, including 20 Hz. Optionally, the shock wave energy generator applies an alternating current to the electrodes to induce a change in the polarity of the electrodes.

According to some aspects, a method for emitting shock waves in a body lumen comprises: positioning a catheter adjacent to an occlusion in a vessel, the catheter comprising a plurality of shock wave emitters disposed at a distal end of a catheter body; and emitting a first plurality of shock waves from the plurality of shock wave emitters in a distal direction so that the shock waves constructively interfere distally of the distal end of the catheter. Optionally, the method comprises advancing the catheter further into the vessel; and emitting a second plurality of shock waves from the plurality of shock wave emitters so that the shock waves constructively interfere at a location that is distal of the first location. Optionally, the method comprises removing a pacemaker lead from the vessel.

According to some aspects, a catheter for use in a body lumen comprises: a catheter body; and at least three shock wave emitters disposed at a distal end of the catheter body, wherein at least a first shock wave emitter of the at least three shock wave emitters can be driven independently of at least a second shock wave emitter of the at least three shock wave emitters. Optionally, the at least three shock wave emitters are arranged such that shock waves emitted from the at least three shock wave emitters can constructively interfere distally of the catheter body. Optionally, the first shock wave emitter of the at least three shock wave emitters comprises an exposed distal tip of a first insulated wire and an exposed distal tip of a second insulated wire. Optionally, the first insulated wire extends along the length of the catheter body and is positioned within a lumen of the catheter body. Optionally, the first shock wave emitter of the at least three shock wave emitters comprises an exposed distal end of an insulated wire and a conductive emitter band that is separated from the exposed distal end of the insulated wire by a spark gap. Optionally, at least three shock wave emitters are arrayed symmetrically about a longitudinal axis of the catheter body. Optionally, the catheter comprises an enclosure positioned to cover the at least three shock wave emitters at the distal end of the catheter body. Optionally, the enclosure is configured to be filled with a conductive fluid. Optionally, the catheter includes a central lumen extending from the proximal end of the catheter to the distal end of the catheter. Optionally, the central lumen is configured to receive a guide wire Optionally, the central lumen is configured to receive a pacemaker wire lead. Optionally, the catheter body comprises an aspiration lumen. Optionally, the at least three shock wave emitters are respectively spaced apart from one another by a distance of between 1 mm and 10 mm.

According to some aspects, a catheter for use in a body lumen comprises: a catheter body; at least three shock wave emitters disposed at a distal end of the catheter body, each shock wave emitter configured to generate a shock wave that propagates distally of the catheter body; and an enclosure that surrounds the at least three shock wave emitters, wherein shock waves are transmitted through the enclosure. Optionally, at least one shock wave emitter of the at least three shock wave emitters comprises an exposed distal tip of a first insulated wire and an exposed distal tip of a second insulated wire. Optionally, at least one shock wave emitter of the at least three shock wave emitters comprises an exposed distal end of an insulated wire and a conductive emitter band that is separated from the exposed distal end of the insulated wire by a spark gap. Optionally, the enclosure is configured to be filled with a conductive fluid. Optionally, the at least three shock wave emitters are arranged such that shock waves emitted from the at least three shock wave emitters can constructively interfere distally of the catheter body. Optionally, the first shock wave emitter of the at least three shock wave emitters comprises an exposed distal tip of a first insulated wire and an exposed distal tip of a second insulated wire. Optionally, the first shock wave emitter of the at least three shock wave emitters comprises an exposed distal end of an insulated wire and a conductive emitter band that is separated from the exposed distal end of the insulated wire by a spark gap. Optionally, the catheter comprises a central lumen extending from a proximal end of the catheter to the distal end of the catheter. Optionally, the central lumen is configured to receive a guide wire. Optionally, the central lumen is configured to receive a pacemaker wire lead. Optionally, the catheter body comprises an aspiration lumen. Optionally, the at least three shock wave emitters are respectively spaced apart from one another by a distance of between 1 mm and 10 mm.

According to some aspects, a method for removing a pacemaker lead comprises: advancing a catheter along the pacemaker lead to a target site comprising fibrotic tissue, the catheter comprising a plurality of shock wave emitters disposed at a distal end of the catheter body, each shock wave emitter configured to generate a shock wave that propagates distally of the catheter body, and wherein at least one shock wave emitter of the plurality of shock wave emitters comprises electrodes separated by a spark gap and at least one electrical connection to an electrode of at least one other shock wave emitter of the plurality of shock wave emitters; and generating one or more shock waves to at least partially break up the fibrotic tissue so that the pacemaker lead can be removed. Optionally, the target site is within the heart. Optionally, the fibrotic tissue is located distally of the distal end of the catheter. Optionally, the pacemaker lead is inserted into a lumen of the catheter to guide the catheter to the target site. Optionally, the method comprises advancing the catheter further along the pacemaker lead to the target site; and generating one or more additional shock waves. Optionally, the catheter further comprises a plurality of shock wave emitters positioned proximally of the distal end of the catheter body, wherein the second plurality of shock wave emitters are respectively configured to generate shock waves that propagate radially from the catheter body; and wherein the method further comprises: generating a plurality of shock waves that propagate radially of the catheter using the second plurality of shock wave emitters to at least partially break up a calcified region of vasculature leading to the target site.

According to some aspects, a catheter for use in a body lumen comprises: a catheter body comprising a chamber, wherein one or more surfaces that define the chamber are formed of a material that at least partially reflects shock waves; and a plurality of shock wave emitters positioned at least partially within the chamber, wherein each shock wave emitter is configured to generate a shock wave, wherein shock waves generated by the plurality of shock wave emitters are at least partially reflected radially outward from the catheter body through an opening in the chamber. Optionally, each of the plurality of shock wave emitters is positioned adjacent to a surface of the one or more surfaces within the chamber, wherein the surface is configured to reflect shock waves radially outward from the surface. Optionally, at least three surfaces of the chamber are formed of the material that at least partially reflects shock waves. Optionally, the catheter comprises an enclosure positioned to at least partially circumscribe the plurality of shock wave emitters, wherein the enclosure defines an outer diameter of the chamber and is configured to facilitate transmission of shock waves from the chamber to a target treatment site. Optionally, the enclosure is positioned to cover the opening in the chamber. Optionally, the enclosure is configured to be filled with a conductive fluid. Optionally, the shock wave emitters are positioned coplanar with a virtual plane that is perpendicular to a longitudinal axis of the catheter.

According to some aspects, a catheter for use in a body lumen comprises: a catheter body; a first plurality of shock wave emitters disposed at a distal end of the catheter body, each shock wave emitter configured to generate a shock wave that propagates distally of the catheter body; and a second plurality of shock wave emitters positioned proximally of the distal end of the catheter body, wherein the second plurality of shock wave emitters are respectively configured to generate shock waves that propagate radially from the catheter body. Optionally, the catheter comprises an enclosure that surrounds at least the first plurality of shock wave emitters, wherein shock waves generated by the first plurality of shock wave emitters are transmitted through the at least one enclosure. Optionally, the first plurality of shock wave emitters is configured to generate a first plurality of shock waves independently of the second plurality of shock wave emitters. Optionally, the first plurality of shock wave emitters is configured to generate a plurality of shock waves simultaneously with the second plurality of shock wave emitters. Optionally, the second plurality of shock wave emitters are positioned at least partially within a chamber of the catheter body, wherein each shock wave emitter of the second plurality of shock wave emitters is configured to generate a shock wave, wherein shock waves generated by the plurality of shock wave emitters are at least partially reflected radially outward from the catheter body through an opening in the chamber. Optionally, one or more surfaces that define the chamber are formed of a material that at least partially reflects shock waves. Optionally, the catheter comprises an enclosure positioned to at least partially circumscribe the second plurality of shock wave emitters, wherein the enclosure defines an outer diameter of the chamber and is configured to facilitate transmission of shock waves from the chamber to a target treatment site. Optionally, at least a first shock wave emitter of the first plurality of shock wave emitters can be driven independently of at least a second shock wave emitter of the first plurality of shock wave emitters. Optionally, at least a first shock wave emitter of the second plurality of shock wave emitters can be driven independently of at least a second shock wave emitter of the second plurality of shock wave emitters. Optionally, at least one shock wave emitter of the first plurality of shock wave emitters can be driven independently of at least one shock wave emitter of the second plurality of shock wave emitters. Optionally, the first plurality of shock wave emitters comprises at least three shock wave emitters. Optionally, the second plurality of shock wave emitters comprises at least three shock wave emitters. Optionally, the first plurality of shock wave emitters and the second plurality of shock wave emitters both comprise at least three shock wave emitters.

BRIEF DESCRIPTION OF THE FIGURES

The invention will now be described, by way of example only, with reference to the accompanying drawings, in which:

FIG. **1**A illustrates an isometric view and a side view of a catheter including a plurality of shock wave emitters, according to some embodiments.

FIG. **1**B illustrates a detailed isometric view of a distal end of the catheter of FIG. **1**A, according to some embodiments.

FIG. **1**C illustrates a detailed isometric view of a shock wave emitter of the catheter of FIG. **1**A, according to some embodiments.

FIG. **1**D illustrates a detailed side view of a distal end of the catheter of FIG. **1**A, according to some embodiments.

FIG. **2**A illustrates a side view of a distal end of a catheter including a plurality of shock wave emitters encased in an enclosure, according to some embodiments.

FIG. **2**B illustrates an isometric view of a distal end of a catheter including a plurality of shock wave emitters encased in an enclosure, according to some embodiments.

FIG. **3**A illustrates an isometric view of a distal end of a catheter including a plurality of shock wave emitters, according to some embodiments.

FIG. **3**B illustrates a side view of a distal end of a catheter including a plurality of shock wave emitters, according to some embodiments.

FIG. **4**A illustrates a front view of a catheter including a plurality of shock wave emitters, according to some embodiments.

FIG. **4**B illustrates a front view of a catheter including a plurality of shock wave emitters, according to some embodiments.

FIG. **4**C illustrates a side view of a catheter including a plurality of shock wave emitters, according to some embodiments.

FIG. **4**D illustrates a side view of a catheter including a plurality of shock wave emitters, according to some embodiments.

FIG. **4**E illustrates a side view of a catheter including a plurality of shock wave emitters encased in an enclosure, according to some embodiments.

FIG. **5** illustrates a system for generating shock waves, according to some embodiments.

FIG. **6** illustrates a method for emitting shock waves in a vessel, according to some embodiments.

FIGS. **7**A-**7**B illustrate the effect of various embodiments of an enclosure such as a cap or angioplasty balloon enclosing shock wave emitters on shock wave propagation.

FIGS. **8**A-**8**C illustrate directional control of shock waves using different shock wave emitter and catheter body configurations.

FIG. **9** illustrate an exemplary catheter including a plurality of shock wave emitters and a lumen for removing a pacemaker lead.

FIG. **10** illustrates an exemplary catheter including shock wave emitters configured to be fired independently from a plurality of shock wave emitters configured to be fired simultaneously.

FIGS. **11**A-**11**C illustrate directional control and constructive interference of shock waves using different shock wave emitter and catheter body configurations.

FIGS. **12**A-**12**C illustrate catheters including a plurality of shock wave emitters, according to some embodiments.

FIGS. **13**A and **13**B illustrate a catheter including forward firing and side firing shock wave emitters according to some embodiments.

FIG. **14** illustrates a method for pacemaker lead removal according to some embodiments.

FIG. **15** illustrates a front view of an exemplary catheter **1500** for generating shock waves according to some embodiments.

DETAILED DESCRIPTION

The following description is presented to enable a person of ordinary skill in the art to make and use the various embodiments and aspects thereof disclosed herein. Descriptions of specific devices, assemblies, techniques, and applications are provided only as examples. Various modifications to the examples described herein will be readily apparent to those of ordinary skill in the art, and the general principles described herein may be applied to other examples and applications without departing from the spirit and scope of the various embodiments and aspects thereof. Thus, the various embodiments and aspects thereof are not intended to be limited to the examples described herein and shown but are to be accorded the scope consistent with the claims.

Efforts have been made to improve the design of electrode assemblies included in shock wave and directed cavitation catheters. For instance, low-profile electrode assemblies have been developed that reduce the crossing profile of a catheter and allow the catheter to more easily navigate calcified vessels to deliver shock waves in more severely occluded regions of vasculature. Examples of low-profile electrode designs can be found in U.S. Pat. Nos. 8,888,788, 9,433,428, and 10,709,462, and in U.S. Publication No. 2021/0085383 all of which are incorporated herein by reference. Other catheter designs have improved the delivery of shock waves, for instance, by specific electrode construction and configuration thereby directing shock waves in a forward direction to break up tighter and harder-to-cross occlusions in vasculature. Examples of forward-firing catheter designs can be found in U.S. Pat. Nos. 10,966,737, 11,478,261, and 11,596,423 and U.S. Publication Nos. 2023/0107690 and 2023/0165598, all of which are incorporated herein by reference.

Described herein are devices, systems, and methods for generating shock waves that propagate in a substantially forward direction, which can be used to treat vascular diseases, such as Chronic Total Occlusion (CTO), Mitral Annular Calcification (MAC), or circumferential calcium, or to treat urinary diseases, such as concretions or kidney stones in the ureter. In accordance with the present disclosure, a catheter includes a catheter body and a plurality of shock wave emitters positioned at a distal end of the catheter body. In some embodiments, each shock wave emitter includes at least two electrodes separated by a spark gap. The electrodes are positioned such that when a high voltage is applied across the electrodes, a shock wave is generated that propagates in a direction forward of the distal end of the catheter body. Shock waves generated by the multiple emitters may constructively interfere forward of the distal end of the catheter body, amplifying the compressive force of the shock waves for treating calcifications forward of the catheter body.

In some embodiments, a plurality of emitters are disposed at the distal end of the catheter. An emitter may include a pair of electrodes that are formed by the exposed distal ends of two wires. In some embodiments, one of the two wires of a first emitter extends to a second emitter to form an electrode of the second emitter. A third wire may have an exposed distal end at the second emitter that forms an electrode pair with the exposed distal end of the wire extending from the first emitter. The third wire may also extend to a third emitter to form part of an electrode pair at the third emitter, and so on for additional emitters. The wires extending between each of the respective emitters may be routed through lumens in the catheter body extending from one emitter to the next.

In some embodiments, an emitter includes a first electrode formed by an exposed conductive end of a wire and a second electrode formed by a conductive band surrounding the end of the wire. The end of the wire is spaced from the conductive band by a spark gap. A third electrode formed by an end of a second wire may be connected to both the first conductive band and a second conductive band to transfer electrical current between the two bands. The second conductive band may form part of an electrode pair with an end of a third wire, thus forming the next emitter in a series of emitters. Any number of emitters may be connected in series by extending wires between conductive bands to transfer an electrical current between each respective band.

In some embodiments, a wire of one of the electrodes at an emitter extends along the catheter body toward a negative terminal of a voltage source, and a wire of an electrode at a different emitter extends along the catheter body toward a positive terminal of the voltage source. Accordingly, plurality of shock wave emitters may be connected in series to a voltage generator via the two wires such that when voltage pulses are applied across the wires at the negative and positive terminal, shock waves are emitted from each of the respective shock wave emitters.

In any of the emitter configurations described herein, voltage polarity (i.e., direction of current flow) may be switched between voltage pulses. Such polarity switching may promote more uniform wear of electrodes and extend device longevity.

In some embodiments, at least one of the shock wave emitters can be driven independently of at least one other shock wave emitter. For instance, one or more emitters may include an electrode pair formed from the exposed tips of two wires, and each of the wires may extend along the length of the catheter body to connect to a respective negative and positive terminal at a voltage source. Additionally, or alternatively, one or more emitters may include an electrode pair formed from an exposed end of a first insulated wire separated by a spark gap from a conductive band. An exposed end of a second insulated wire may be connected to the emitter band, and both the first and second insulated wires may extend along the length of the catheter body to connect to a respective negative and positive terminal at a voltage source. When a voltage is applied across the two wires connected to the voltage source, a shock wave can be generated at the respective shock wave emitter without producing shock waves at any other emitters provided on the catheter. Accordingly, in some instances, the design is such that each electrode pair can spark separately from the other electrode pairs, including adjacent electrode pairs. The emitters may be driven sequentially, for instance, in a clockwise and or counter-clockwise manner.

In some embodiments, the shock wave emitters are enclosed within an enclosure such as a fluid filled cap or balloon. The cap or balloon may mitigate thermal injury to soft tissue and reduce cavitation stresses by limiting expansion of the vapor bubbles produced during shock wave generation to the interior of the cap. For instance, the vapor bubbles hit the enclosure wall before reaching their maximum potential size, thus inducing collapse, and reducing cavitation stress and preventing soft tissue injury that can be caused by tensile stresses during cavitation bubble collapse. As described further below, the shape of the enclosure may also impact the form of the shock waves and vapor bubbles as well as the manner in which these shock waves and vapor bubbles propagate forward of the catheter. In some embodiments, the shock wave emitters are exposed at the distal end of the catheter and cavitation bubble formation and collapse on the surface of the target calcification further contributes to fragmentation of the calcification.

Shock wave emitters may be connected to a voltage pulse generator capable of applying high voltage, high frequency electrical pulses to simultaneously generate a plurality of shock waves from the plurality of shock wave emitters that can propagate forward of the distal end of the catheter to impinge on a treatment area positioned distal of the catheter. In some embodiments, the catheters described herein may be connected to a pump in communication with a fluid source for injecting a conductive fluid such as saline and/or contrast solution into the catheter. In some embodiments, the conductive fluid is injected into an enclosure that encloses the plurality of shock wave emitters, as described above.

In some embodiments, the catheters described herein can be inserted into a body lumen, such as a blood vessel, for instance, to treat a buildup of calcification. The catheter may be advanced within the body lumen until the distal end of the catheter reaches a desired distance from the target treatment area. Once in position, a plurality of shock waves may be emitted from the plurality of shock wave emitters such that the shock waves propagate in a unison direction toward the target treatment area. In some embodiments, the shock waves constructively interfere with one another distally of the distal end of the catheter, thus compounding the peak compressive force of the plurality of shock waves relative to each of the individual shock waves emitted by the respective shock wave emitters. In some embodiments, after breaking up a portion of the target treatment area (e.g., a portion of the calcification/occlusion), the catheter may be advanced further into the vessel toward a second target treatment area (e.g., a newly exposed portion of the occlusion/calcification), and a second plurality of shock waves may be emitted targeting the treatment area. This process may be iterated any number of times until the occlusion/calcification has been successfully treated.

As used herein, the term "electrode" refers to an electrically conducting element (typically made of metal) that receives electrical current and subsequently releases the electrical current to another electrically conducting element. In the context of the present disclosure, electrodes are often positioned relative to each other, such as in an arrangement of an inner electrode and an outer electrode. Accordingly, as used herein, the term "electrode pair" refers to two electrodes that are positioned adjacent to each other such that application of a sufficiently high voltage to the electrode pair will cause an electrical current to transmit across the gap (also referred to as a "spark gap") between the two electrodes (e.g., from an inner electrode to an outer electrode, or vice versa, optionally with the electricity passing through a conductive fluid or gas therebetween). In some contexts, one or more electrode pairs may also be referred to as an electrode assembly. In the context of the present disclosure, the term "emitter" broadly refers to the region of an electrode assembly where the current transmits across the electrode pair, generating a shock wave. The terms "emitter sheath" and "emitter band" refers to a continuous or discontinuous band of conductive material that may form one or more electrodes of one or more electrode pairs, thereby forming a location of one or more emitters.

In some embodiments, the catheters described herein may be an IVL catheter. In some embodiments, an IVL catheter may be a so-called "rapid exchange-type" ("Rx") catheter provided with an opening portion through which a guide wire can be guided (e.g., through a middle portion of a central tube in a longitudinal direction). In other embodiments, an IVL catheter may be an "over-the-wire-type" ("OTW") catheter in which a guide wire lumen is formed throughout the overall length of the catheter, and a guide wire is guided through the proximal end of a hub.

Although shock wave devices described herein generate shock waves based on high voltage applied to electrodes, it should be understood that a shock wave device additionally or alternatively may comprise a laser and optical fibers as a shock wave emitter system whereby the laser source delivers energy through an optical fiber and into a fluid to form shock waves and cavitation bubbles.

In the following description of the disclosure and embodiments, reference is made to the accompanying drawings in which are shown, by way of illustration, specific embodiments that can be practiced. It is to be understood that other embodiments and examples can be practiced and changes can be made without departing from the scope of the disclosure.

In addition, it is also to be understood that the singular forms "a," "an," and "the" used in the following description are intended to include the plural forms as well, unless the context clearly indicates otherwise. It is also to be understood that the term "and/or" as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It is further to be understood that the terms "includes, "including," "comprises," and/or "comprising," when used herein, specify the presence of stated features, integers, steps, operations, elements, components, and/or units, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, units, and/or groups thereof.

FIGS. **1A-1D** illustrate an exemplary catheter **100** for generating shock waves in a forward direction, according to some embodiments. The catheter **100** may be used to fragment, crack, or otherwise break up calculi within the human body, for instance, to treat various occlusions within blood vessels, kidney stones in the ureter, and so on.

FIG. **1A** illustrates an isometric view and a side view of a distal portion of a catheter **100**. The catheter **100** of FIG. **1A** includes a catheter body **101** with a distal end **104**. A plurality of shock wave emitters **106**, **108**, and **110** (shown more clearly in FIG. **1B**) are positioned at the distal end **104** of the catheter body **101**. Each shock wave emitter **106-110** is configured to generate a shock wave that propagates distally of the catheter body **101** (i.e., distally of distal end **104**). In some embodiments, the catheter body **101** may have an outer diameter of between 3 Fr and 14 Fr (French Gauge). In some embodiments, the catheter body **101** may have an outer diameter of between 1 Fr and 20 Fr. In some embodiments, the catheter body **101** may have an outer diameter of between 1 Fr and 100 Fr. In some embodiments, the catheter body **101** may have an outer diameter of at least 1 Fr, at least 2 Fr, at least 3 Fr, at least 4 Fr, at least 5 Fr, at least 6 Fr, at least 7 Fr at least 8 Fr, at least 9 Fr, at least 10 Fr, at least 11 Fr, at least 12 Fr, at least 13 Fr, at least 14 Fr, at least 15 Fr, at least 16 Fr, at least 17 Fr, at least 18 Fr, at least 19 Fr, or at least 20 Fr. In some embodiments, the catheter body may have an outer diameter of no more than 20 Fr, no more than 19 Fr, no more than 18 Fr, no more than 17 Fr, no more than 16 Fr, no more than 15 Fr, no more than 14 Fr, no more than 13 Fr, no more than 12 Fr, no more than 11 Fr, no more than 10 Fr, no more than 9 Fr, no more than 8 Fr, no more than 7 Fr, no more than 6 Fr, no more than 5 Fr, no more than 4 Fr, no more than 3 Fr, no more than 2 Fr, or no more than 1 Fr.

When a sufficiently high voltage (pulse) is applied across two electrodes at each shock wave emitter, a shock wave is generated at each shock wave emitter as electrical current flows from the first electrode to the second electrode, resulting in a plurality of shock waves from the plurality of shock wave emitters. The shock wave emitters **106-110** may be arranged such that shock waves emitted from the plurality of shock wave emitters **106-110** to constructively interfere distally of the catheter body **101**. Thus, the positioning of the shock wave emitters **106-110** can maximize the shock wave intensity distally of the catheter body by causing shock waves emitted by each respective emitter to combine with one another to produce an amplified combined shock wave, for instance as illustrated in FIGS. **11**A-**11**C.

FIG. **1**B illustrates a detailed isometric view of the distal end **104** of catheter **100**. In the illustrated embodiment, the shock wave emitters **106**, **108**, and **110** are evenly spaced (positioned at increments of about 120 degrees) about the longitudinal axis **181**; however, as described throughout the specification, a variety of different spacing configurations can be implemented without deviating from the scope of the disclosure. In some embodiments, the shock wave emitters may be spaced apart from one another by a distance of between 0.1 mm and 20 mm. In some embodiments, the shock wave emitters may be spaced apart from one another by a distance of between 1 mm and 10 mm. In some embodiments, the shock wave emitters may be spaced apart from one another by between 2 mm and 5 mm. The shock wave emitters may be spaced apart from one another by at least 1 mm, at least 2 mm, at least 3 mm, at least 4 mm, at least 5 mm, at least 6 mm, at least 7 mm, at least 8 mm, at least 9 mm, at least 10 mm, at least 12 mm, at least 13 mm, at least 14 mm, at least 15 mm, at least 16 mm, at least 17 mm, at least 18 mm, at least 19 mm, or at least 20 mm. The shock wave emitters may be spaced apart from one another by no more than 20 mm, no more than 19 mm, no more than 18 mm, no more than 17 mm, no more than 16 mm, no more than 15 mm, no more than 14 mm, no more than 13 mm, no more than 12 mm, no more than 11 mm, no more than 10 mm, no more than 9 mm, no more than 8 mm, no more than 7 mm, no more than 6 mm, no more than 5 mm, no more than 4 mm, no more than 3 mm, no more than 2 mm, or no more than 1 mm. The emitters may be spaced apart from one another by a distance set to optimize the constructive interference of shock waves generated by the emitters (e.g., depending on sonic output from individual emitters, acoustic properties of the propagating medium, etc.) In some embodiments, the distance between shock wave emitters is the distance between the two center points of two respective electrode pairs. In some embodiments, the distance between shock wave emitters is measured as the distance between the center points of two respective emitter bands.

In the exemplary embodiment of FIG. **1**B, a first shock wave emitter **106** of the plurality of shock wave emitters includes a distal tip **113** of a first insulated wire **112**. The insulated wire **112** extends along the length of the catheter body **101** from the distal end **104** (e.g., so that it can be connected to a voltage source proximally of the distal end (for instance, at a proximal end of the catheter), as described further below). A second insulated wire **114** extends from the first shock wave emitter **106** to the second shock wave emitter **108**. The second insulated wire includes a first exposed distal tip **115***a* forming an electrode pair with distal tip **113** separated by a spark gap, thus forming shock wave emitter **106**, and a second exposed distal tip **115***b* forming part of an electrode pair at shock wave emitter **108**, as described below. As used herein, an "exposed end," "exposed tip," and/or "exposed distal tip" of an insulated wire may refer to a portion of the wire from which the insulation has been removed, thus revealing a portion of the conductive wire. However, while the emitters herein are often described as including the exposed distal ends/tips of insulated wires, it should be understood that any suitable conductor may serve as an electrode of the emitters.

The second insulated wire **114** extends proximally from shock wave emitter **106** into the catheter body **101** for a first distance, and loops around, for instance as illustrated by the bend **195** forming the U-shaped portion of insulated wire **114**, to extend distally toward shock wave emitter **108**. A third insulated wire **116** includes a first exposed distal tip **117***a* at shock wave emitter **108**. The second exposed distal tip **115***b* of second insulated wire **114** and first exposed distal tip **117***b* of the third insulated wire **116** form an electrode pair separated by a spark gap, thus forming shock wave emitter **108**. The third insulated wire **116** wire extends from the second shock wave emitter **108** to a third shock wave emitter **110**. Similar to the second insulated wire **114**, the third insulated wire **116** extends proximally into the catheter body **101** for a first distance, and loops around to extend distally toward shock wave emitter **110**. The third insulated wire **116** includes a second exposed distal tip **117***b* at shock wave emitter **110**, forming an electrode pair with exposed distal tip **119** of a fourth insulated wire. The exposed distal tips **117***b* and **119** form an electrode pair separated by a spark gap, thus forming third shock wave emitter **110**. The fourth insulated wire **118** extends proximally into the catheter body and along the length of the catheter body **101** from the distal end **104** to connect to a positive terminal of a voltage source. Accordingly, when a voltage is applied across the first insulated wire **112** connected to the negative terminal of the voltage source and the fourth insulated wire connected to the positive terminal of the voltage source, a plurality of shock waves are simultaneously generated as an electrical current traverses the spark gaps separating the exposed distal tips of each insulated wire at shock wave emitters **106-110**.

In some embodiments, the shock wave emitters **106-110** of the catheter **100** shown in FIG. **1**B are electrically connected in series such that an electrical pulse applied across insulated wires connected to a negative and positive terminal of a voltage source (such as wire **112** and **118**), respectively, causes each of the plurality of shock wave emitters to emit a respective shock wave. In some embodiments, at least one first shock wave emitter of a plurality of shock wave emitters can be driven independently of at least a second shock wave emitter of the plurality of shock wave emitters. Accordingly, in some embodiments, rather than extending wires between all of the shock wave emitters such that applying a single voltage pulse causes each of the shock wave emitters to generate shock waves in series, one or more shock wave emitters can each include an electrode pair configured to generate shock waves independently of the other shock wave emitters. In some embodiments, the electrode pair at each shock wave emitter (e.g., shock wave emitters **106-110**) can be formed of the exposed distal tips of a first and second wire that each extend along the length of catheter **100** from the distal end **104** to electrically couple to a respective positive and negative terminal (or to ground) of a voltage source (i.e., each shock wave emitter may be connected to a respective channel of a relay such that it can be driven independently of the other emitters). In such embodiments, when a voltage pulse is applied across the first and second wire of an independently driven shock wave emitter, a current flows from an exposed distal tip of the first insulated wire to the exposed distal tip of the second insulted wire to generate a shock wave, but that shock wave emitter is electrically isolated from the remaining shock wave emitters.

In some embodiments, catheter body **101** includes one or more lumens extending within the catheter body. In some embodiments, one or more of the insulated wires (e.g., wires **112** and **118** in FIG. **1**B) extend along the length of the catheter within a respective lumen to connect to a voltage source. As described above, other insulated wires (e.g., wires **114** and **116** of FIG. **1**B) are routed between respective shock wave emitters to carry the current received from the voltage between each of the emitters. Accordingly, the wires routed between respective shock wave emitters may extend into a first lumen of the catheter body **101** in a first direction toward a first shock wave emitter and extend into a second lumen in a second direction toward a second shock wave emitter. As depicted in FIG. **1**B, insulated wire **114** is formed into a U-shape, where the parallel portions of the U-shape respectively of extend into lumens **140** and **142**. The bend **195** in the U-shape of wire **114** is formed within a cavity **180** formed between a surface **161** of a first section **160** of catheter body **101** and second surface **163** of a second section **162** of the catheter body **101** near the distal end **104**. In some embodiments, lumens **140**, **142**, and/or **144** extend from respective orifices in the distal most surface of catheter **100** at distal end **104** into section **160** of the catheter body **101** to a respective orifice of surface **161** facing cavity **180**. In some embodiments, any of the respective lumens **140**, **142**, and/or **144** extend into the second section **162** of catheter body **101** from a respective orifice of surface **163** on the opposite side of cavity **180** along the same respective longitudinal axes as in section **162**.

In some embodiments, cavity **180** is formed into section **160** of catheter body **101**. Section **160** may be a removable tip that can be friction fit onto section **162**. Cavity **180** may be a hollow portion of section **160** that is configured such that a portion **161** of section **162** can extend into the cavity **180** when section **160** is friction fit with section **162** (e.g., such that sections **160** and **162** overlap with one another when section **160** is friction fit to section **162**). The removability of section **160** from section **162** can allow for placement/replacement of wires and/or other device maintenance.

In some embodiments, the catheter **100** includes a central lumen **150** extending from the distal end **104** of the catheter along the length of the catheter body **101**. In some embodiments the central lumen **150** may be enclosed within a tube **152** that extends from section **160** to section **162** through cavity **180** within the catheter body **101**, as shown in FIG. **1**D. In some embodiments, the central lumen **150** extends from an orifice at the distal end **104** to an orifice at the opposite end of catheter body **101**. The central lumen **150** may be configured to receive a guide wire. For instance, the guide wire can be inserted into the catheter **100** via central lumen **150** proximally of the distal end **104** and exit the catheter via the central lumen **150** at the distal end **104**. The guide wire can be used to guide the catheter into place within a body lumen (e.g., blood vessel, urinary tract, or other organ).

In some embodiments, the central lumen **150** is configured to receive a pacemaker wire lead at the distal end **104** of the catheter body **101** for removing the pacemaker wire lead from a tissue, such as cardiac tissue (for instance, as shown in FIG. **9**). Pacemaker leads can be difficult to remove due to dense calcification and/or fibrotic tissue build-up. This calcification build-up can make extraction more difficult for the physician and riskier for the patient. The catheter described herein can be used to first break-up these dense calcifications using the shock waves generated by the plurality of shock wave emitters before removing the pacemaker lead. Breaking up the calcifications prior to removing the leads can lead to dramatic reduction in removal time.

In some embodiments, the catheter body **101** includes an aspiration lumen. In some embodiments, the aspiration lumen is for removing debris from a body lumen. In some embodiments central lumen **150** can be configured for aspiration. In some embodiments, the catheter body may include a separate lumen in addition to the central lumen **150** for aspiration. In some embodiments, the catheter **100** includes a marker band for determining an orientation of the catheter within a body lumen.

FIG. **1**C illustrates a detail view of shock wave emitter **110**, in accordance with some embodiments. Shock wave emitter **110** includes insulated wire **116** and insulated wire **118** described above with reference to FIG. **1**B. Insulated wire **116** includes a conductive wire **130** disposed within an insulating tube **132**. In some embodiments, the conductive wire **130** is a copper wire and insulating tube **132** is a polyimide tube. Similarly, insulated wire **118** includes a conductive wire **134** disposed within an insulating tube **136**. In some embodiments, the conductive wire **134** is a Molybdenum wire and the insulating tube **136** is also a polyimide tube. In some embodiments, at least a portion of both of the insulating tubes **132** and **136** are disposed within a single outer insulating tube **138**. In some embodiments, the outer insulating tube **138** is also a polyimide tube that provides an additional layer of insulation. The outer insulating tube **138** may separate the two insulated wires **116** and **118** from one another (e.g., by surrounding a portion of each of the insulated wires individually), and in turn separate their respective exposed distal tips (**117***b* and **119**) by a spark gap. An outer insulating tube **138** may be positioned at each of shock wave emitters **106**, **108**, and **110** to provide an additional layer of insulation and to maintain a spark gap between the exposed distal tips (e.g., **117***a*, **115***b*, **115***a*, and **113**). As described above, the distal tips **117***b* and **119** of conductive wires **130** and **134**, respectively, are exposed at the distal end of the catheter **100** to allow for generation of shock waves. The exposed distal tips **117***b* and **119** form an electrode pair separated by a spark gap such that when a voltage is applied across the conductive wires **130** and **134**, a current flows from exposed distal tip **117***b* to the exposed distal tip **119** to generate a shock wave.

In some embodiments, the shock wave emitters **106-110** are arrayed symmetrically about the longitudinal axis **181** of the catheter body, for instance, as shown in FIG. **1**B. In some embodiments, the shock wave emitters are instead arrayed asymmetrically about the longitudinal axis of the catheter body. For example, two of the shock wave emitters may be positioned more closely to one another than a third shock wave emitter. Symmetric arrangement of the shock wave emitters about the longitudinal axis may be desirable for optimizing constructive interference between all of the shock wave emitters. However, arranging the shock wave emitters asymmetrically about the axis can provide for asymmetric shock waves, which may be beneficial, for instance, if an occlusion is concentrated at various locations within a body lumen relative to the distal end **104** of the catheter **100** (i.e., if the occlusion is more concentrated at various location about the circumference of the catheter **100**).

In some embodiments, a distal most surface **182** of one or more shock wave emitters of the plurality of shock wave emitters is flush with a distal most surface **183** of the distal end of the catheter body, for instance, as shown in FIG. **1**D. In some embodiments, a distal most surface **182** of a shock wave emitter of the plurality of shock wave emitters is instead positioned forward from a distal most surface **183** of the distal end of the catheter body **101**, for instance, as shown in FIG. **4**D. Such a configuration may be preferable, for instance, if direct contact between the shock wave emitters and an occlusion is desired. Alternatively, the distal most surface of any of the shock wave emitters may be recessed from a distal most surface of the distal end of the catheter body, also as shown in FIG. **4**D. Such a design be preferred over the flush configuration if a user desired to prevent direct contact between the shock wave emitters and the occlusion/calcification. Additionally, when the distal most surfaces of the shock wave emitters are recessed from the distal most surface of the distal end of the catheter body, the resulting cavitation after pulsing the shock wave emitters may travel further distally of the catheter body relative to when the distal most end of the emitters is positioned flush with the distal most end of the catheter body and/or may produce a greater sonic output relative to when the distal most end of the emitters is positioned flush with the distal most end of the catheter body.

In some embodiments, the plurality of shock wave emitters (e.g., **106-110**) are arrayed about a longitudinal axis of the catheter body at the same distal location relative to the distal end of the catheter body, for instance, as shown in FIG. **1**D. In some embodiments, a first shock wave emitter of the plurality of shock wave emitters is positioned at a first location relative to the distal end of the catheter body and a second shock wave emitter of the plurality of shock wave emitters is positioned at a second location relative to the distal end of the catheter body. For example, a first shock wave emitter may be positioned such that its distal most surface is flush with the distal end **104** of the catheter body **101**, one or more of the shock wave emitters may be positioned distally relative to the first shock wave emitter, and one or more shock wave emitters may be placed proximally relative to the first shock wave emitter. Alternatively, a first shock wave emitter may be positioned such that its distal most surface is flush with the distal end **104** of the catheter body **101**, a second shock wave emitter may be positioned proximally relative to the first shock wave emitter, and a third shock wave emitter may be positioned proximally relative to the second shock wave emitter. It should be understood that the aforementioned configurations are meant to be exemplary, and the shock wave emitters could be positioned in a variety of different locations relative to one another without deviating from the scope of this disclosure. A variety of different arrangements of the shock wave emitters that may be included on catheter **100** are shown below in FIGS. **4**A-**4**E. Additionally, it should be understood that while catheter **100** is described as including three shock wave emitters, in some embodiments, the catheters described herein may include additional shock wave emitters (i.e., more than the three provided on catheter **100**) positioned at the distal end of the catheter body. The additional shock wave emitters may be included, for instance, to generate a more powerful shock wave for treating occlusions or otherwise breaking up calculi within a human body.

In some embodiments the catheter **100** illustrated in FIGS. **1**A-**1**D includes an enclosure **190** positioned to cover the plurality of shock wave emitters at the distal end **104** of the catheter body **101**, as shown in FIGS. **2**A and **2**B. In some embodiments, the enclosure **190** is a cap, and in some embodiments, the enclosure **190** is an angioplasty balloon. In some embodiments, the enclosure **190** can be filled or inflated by pumping a conductive fluid, such as saline and/or contrast agent into the interior volume of the balloon. When attached, the enclosure may minimize the effect of negative pressures induced by cavitation bubbles that result from shock wave generation. As described above, without the enclosure, cavitation bubbles may exert tensile forces on a target treatment area when they collapse upon impact with the treatment area. However, with enclosure **190** attached and enclosing the shock wave emitters, the cavitation bubbles resulting from shock wave generation are separated from the target treatment area by the enclosure **190**. Accordingly, the enclosure can protect soft tissue from potential damage resulting from the cavitation bubbles.

The enclosure **190** may be filled or inflated before, during, or after applying shock waves to a treatment region. For example, in some embodiments, after forward directed shock waves are initiated using the shock wave emitters **106-110** at the distal end **104** of the catheter **100** to break apart an occlusion, the catheter **100** is advanced further into a patient's vessel, and the enclosure **190** is inflated in the region of the occlusion to further treat the region. The conductive fluid may aid in cooling the device and dissipating heat generated during the formation of vapor bubbles that result from shock wave generation. The enclosure may also shield the shock wave emitters from direct contact with vessel walls and/or direct contact with an occlusion inside a body lumen. In some embodiments the enclosure **190** is formed of a thin, acoustically transparent material, such as polyethylene or nylon, which can provide for efficient fluid-to-tissue transmission and effective coupling of the pressure pulse from shock wave emitter to the occlusion or other calcification. In some embodiments, the catheter **100** may be connected to a fluid source **540** and fluid pump **550** (e.g., as shown in FIG. **5**). The fluid pump **550** may fill the interior volume of the enclosure **190** with fluid to a certain pressure. The conductive fluid may be circulated within the interior volume of the enclosure **190** via a lumen of the catheter **100**, for instance central lumen **150**, by injecting the fluid into the enclosure **190** by a fluid pump **550** shown in FIG. **5** and drawing it into a fluid return line (not shown). In some embodiments, a conductive fluid is replenished (e.g., by continually flowing the conductive fluid into the enclosure **190** via a supply line, such as central lumen **150** or a different supply line, and allowing excess fluid to exit the enclosure **190** via a return line) during shock wave treatment to displace gas bubbles generated during shock wave emission. In some embodiments, instead of or in addition to a fluid pump, conductive fluid is supplied by a fluidically connected pressurized chamber (e.g., a saline bag). The method for replenishing conductive fluid described above may apply equally to any of the enclosures (e.g., **490**, **790**, **1290**) included on any of the catheters described herein.

With respect to catheter **100** described above, the shock wave emitters are configured to respectively emit shock waves by creating a spark across a spark gap formed between the exposed distal ends of two wires. In some embodiments, such as the exemplary embodiment depicted in FIGS. **3**A and **3**B, the shock wave emitters may include conductive emitter bands. Shock waves may be generated by creating a spark across a spark gap between an exposed tip of a wire and a conductive emitter band. In some embodiments, the conductive emitter bands are conductive cylindrical tubes that at least partially circumscribe wires used to either generate shock waves and/or transfer current between multiple conductive emitter bands. Accordingly, in some embodiments, the shock wave emitters of the catheters described herein may include a respective conductive emitter band, a first insulated wire with an exposed distal tip positioned such that a spark gap is formed between the exposed distal tip and the conductive emitter band, and a second insulated wire that is electrically connected (e.g., soldered) to the conductive emitter band. The connected wire can be routed to a next shock wave emitter to transfer the electrical current from the first conductive emitter band to an electrode of the next shock wave emitter.

FIGS. **3**A and **3**B illustrate an exemplary catheter **200** for generating shock waves in a forward direction that includes shock wave emitters formed by conductive emitter bands. Catheter **200** includes a catheter body **201** having a distal end **204**. The catheter **200**, as shown, includes a plurality of shock wave emitters **206-210** disposed at the distal end **204** of the catheter body **201**. Each shock wave emitter **206-210** can be configured to generate a shock wave that propagates distally of the catheter body **201** (i.e., distally of distal end **204**). The shock wave emitters **206-210** can be arrayed about a longitudinal axis of the catheter body **201** and configured such that shock waves emitted from the plurality of shock wave emitters **206-210** to constructively interfere distally of the catheter body **201**. Thus, the positioning of the shock wave emitters **206-210** can maximize the shock wave intensity distally of the catheter body by causing shock waves emitted by each respective emitter to combine with one another to produce an amplified combined shock wave, for instance as illustrated in FIGS. **11**A-**11**C.

FIGS. **3**A and **3**B illustrate an isometric view of the distal end **204** of catheter **200** and a side view of the distal end of catheter **200**, respectively. In some embodiments, the shock wave emitters **206-210** of the catheter **200** shown in FIGS. **3**A and **3**B are electrically connected in series. Shock wave emitter **206** includes a conductive band **270** with a distal end and a proximal end, the distal end of the shock wave emitter **206** positioned relatively closer to the distal end **204** of the catheter body **201**. The conductive band **270** may form an electrode pair with an exposed end **213** of a first insulated wire **212**, the exposed end **213** positioned at the distal end of conductive band **270**. The insulated wire **212** may be positioned within the conductive band **270** and separated by a spark gap from the conductive band **270** for generating shock waves when a voltage is applied across the spark gap. In some embodiments, the insulated wire **212** extends along the length of catheter body **201** from within the conductive band **270** at the distal end **204** such that the first insulated wire **212** can be connected to a voltage source.

In some embodiments, shock wave emitter **206** includes a second insulated wire **214** with a first exposed end **215***a* electrically connected (e.g., may be soldered, crimped, taped, adhered, clamped, or otherwise electrically connected to) to the conductive emitter band **270**. In some embodiments, the insulated wire **214** is disposed at least partially within the interior of conductive emitter band **270** and the exposed end **215***a* is electrically connected (e.g., may be soldered, crimped, tapes, clamped, or otherwise connected to) to an inner surface of the conductive band **270**. It should be understood, however, that the exposed end **215***a* could be electrically connected to any conductive surface of the conductive band **270**. In some embodiments, a second exposed end (not shown) of the second insulated wire **214** is electrically connected to a conductive band **272** of a second shock wave emitter **207** to transfer the voltage between the conductive band **270** of the first shock wave emitter **206** and the conductive band **272** of the second shock wave emitter **207**. Similar to conductive band **270**, the conductive band **272** has a distal end and a proximal end, the distal end of the conductive band **272** positioned relatively closer to the distal end **204** of the catheter body **201**. Shock wave emitter **207** includes a first exposed tip **217***a* of a third insulated wire **216** positioned at least partially within the interior of conductive emitter band **272**, the exposed tip **271***a* positioned at the distal end of the conductive emitter band **272** such that shock waves generated by the shock wave emitter **207** propagate distally. Insulated wire **216** is positioned to form a spark gap between the exposed tip **217***a* and the conductive emitter band **272** such that an electrical current can flow between the conductive emitter band **272** and the exposed tip **217***a* to generate a shock wave distally of the distal end of the catheter body **201**.

Insulated wire **216** is also positioned to transfer current from the second shock wave emitter **207** to a third shock wave emitter **208**, and a spark gap is formed between a second exposed end **217***b* of insulated wire **216** and a conductive band **274** at the distal end of the conductive emitter band, as described below. Specifically, insulated wire **216** is inserted into both conductive bands **272** and **274** from the proximal end of each conductive band such that a portion of insulated wire **216** extends into both conductive bands **272** and **274** toward the distal end of each respective conductive emitter band. Insulated wire **216** includes a second exposed end **217***b* disposed at the distal end of the conductive emitter **274**. Insulated wire **216** is positioned such that a spark gap separates the second exposed end **217***b* from the conductive band **274**. Accordingly, when an electrical current flows between exposed end **217***b* and conductive band **274** another shock wave is generated at shock wave emitter **208** and propagates distally of the shock wave emitter **208**.

In some embodiments, a first exposed end **219***a* of a fourth insulated wire **218** is electrically connected (e.g., may be soldered, crimped, tapes, clamped, or otherwise electrically connected to) to the conductive band **274**. In some embodiments, the insulated wire **218** is positioned at least partially within the interior of conductive band **274** and the exposed end **219***a* is electrically connected (e.g., may be soldered, crimped, tapes, clamped, or otherwise electrically connected to) to an inner surface of the conductive band. As with shock wave emitters **206** and **207**, it should be understood that the exposed end **219***a* could be electrically connected to any conductive surface of the conductive band **274**. In some embodiments, a second exposed end **219***b* of the second insulated wire **218** is electrically connected to a conductive band **276** of a fourth shock wave emitter **209** to transfer the electrical current between the conductive band **274** of the third shock wave emitter **208** and the conductive band **276** of the fourth shock wave emitter **209**. More specifically, similar to conductive band **270**, **272**, and **274**, the conductive band **276** has a distal end and a proximal end, the distal end positioned relatively closer to the distal end **204** of the catheter body **201**. Insulated wire **218** may extend outwardly from the distal end of the conductive emitter band **274** and exit the conductive emitter band **274** from its distal end. Insulated wire **218** may then be directed toward the next shock wave emitter in the series, shock wave emitter **209**. Insulated wire **218** may extend into the distal end of conductive band **276** and a second exposed end **219***b* of insulated wire **218** may be electrically connected to the conductive band **276** to transfer an electrical current between the conductive band **274** of the third shock wave emitter **208** and the conductive band **276** of the fourth shock wave emitter **209**.

Shock wave emitter **209** additionally includes a first exposed end **221***a* of a fifth insulated wire **220**. The fifth insulated wire is disposed at least partially within the interior of conductive band **276**, and the first exposed end **221***a* is disposed at the distal end of the conductive emitter band

276. Insulated wire **220** is positioned within conductive emitter band **276** such that the exposed end **221***a* is separated by a spark gap from the conductive emitter band **276**. Accordingly, when a current flows across the spark gap between the conductive emitter band **276** and the exposed end **221***a* a spark is created thus generating a shock wave that propagates distally of the distal end of the catheter body **201**.

Similar to insulated wire **216**, in some embodiments, the insulated wire **220** is routed to a fifth shock wave emitter **210** and an exposed end **221***b* of wire **220** is separated by a spark gap from a conductive band **278** at the distal end of the conductive emitter band, as described in more detail below. Insulated wire **220** may be inserted into both conductive bands **276** and **278** from the proximal end of each respective conductive band. A portion of insulated wire **220** extends into both conductive bands **276** and **278** toward the distal end of each respective conductive emitter band, for instance, as shown in the side view of catheter **200** in FIG. **3**B. Insulated wire **220** may include a second exposed end **221***b* positioned at the distal end of the conductive emitter band **278** (i.e., the end of the conductive emitter band facing the distal end of catheter body **201**), as described above. The insulated wire **220** may be positioned such that the exposed end **221***b* is separated by a spark gap from the conductive emitter band **278**. Accordingly, when the current flows between the exposed tip **221***b* to conductive emitter band **278** a shock wave is generated by shock wave emitter **210**. A sixth insulated wire **222** with an exposed tip **223** may be electrically connected (e.g., soldered crimped, tapes, clamped, or otherwise electrically connected to) to the conductive emitter band **278** and may extend along the length of the catheter body **201** from the conductive emitter band **278** at the distal end **204** to connect to a positive terminal of a voltage source. Accordingly, when a voltage is applied across the negative and positive terminal, and thus across wires **212** and **222**, a plurality of shock waves are generated at each of the shock wave emitters **206**-**210**.

In some embodiments, each of the respective shock wave emitters **206**-**210** are positioned within a respective cavity **240** formed into the outer circumferential surface of catheter body **201**. In some embodiments, each cavity **240** has a semi-circular shape sized such that a respective conductive emitter **270**-**278** band having a cylindrical shape can be positioned at least partially within a semi-circular cavity **240**. In some embodiments, each cavity **240** extends along the length of catheter body **201** from distal end **204** (e.g., to a proximal end of the catheter body **201**). In some embodiments, one or more of the insulated wires included in shock wave emitters **206**-**210** extend within a respective cavity **240** along the length of the catheter body **201** to connect to a voltage source.

In some embodiments, the catheter **200** includes a central lumen **250** extending from the distal end **204** along the length of the catheter body **201**. The central lumen **250** may be configured to receive a guide wire. For instance, the guide wire can be inserted into the catheter **200** via central lumen **250** proximally of the distal end **204** and exit the catheter via the central lumen **250** at the distal end **204**. The guide wire can be used to guide the catheter into place within a body lumen (e.g., blood vessel or other organ). In some embodiments, the central lumen **250**, like central lumen **150** of catheter **100**, is configured to receive a pacemaker wire lead at the distal end **204** of the catheter body **201** for removing the pacemaker wire lead from a tissue, such as cardiac tissue (for instance, as illustrated in FIG. **9**, below). In some embodiments, the catheter body **201** includes an aspiration lumen. In some embodiments, the aspiration lumen is for removing debris from a body lumen. In some embodiments central lumen **250** can be configured for aspiration. In some embodiments, the catheter body may include a separate lumen (not shown) for aspiration. In some embodiments, the catheter **100** includes a marker band for determining an orientation of the catheter within a body lumen.

Although catheter **200** is described as having five shock wave emitters electrically connected in series such that an electrical pulse applied to a first shock wave emitter of the plurality of shock wave emitters causes each of the plurality of shock wave emitters to emit a respective shock wave, it should be understood that the shock wave emitters could be configured such that any of the shock wave emitters of the plurality of shock wave emitters **206**-**210** can be driven independently of any of the other shock wave emitter of the plurality of shock wave emitters. An exemplary embodiment illustrating a catheter including a first set of shock wave emitters configured to be driven independently of one another and a second set of shock wave emitters configured to be driven in series is illustrated below in FIG. **10**.

In the exemplary embodiment depicted in FIGS. **3**A and **3**B, the plurality of shock wave emitters **206**-**210** of catheter **200** are arrayed symmetrically about the longitudinal axis **281** of the catheter body and a distal most surface of each of shock wave emitters **206**-**210** is positioned such that it is recessed from a distal most surface of the distal end **204** of the catheter body **201**. However, it should be understood that the plurality of shock wave emitters on the catheters described herein (e.g., catheter **100** and catheter **200**) could be arranged in a variety of configurations without deviating from the scope of the claims, for instance, as shown in the exemplary embodiments depicted in FIGS. **4**A-**4**E. It should be understood that any of the features illustrated in FIGS. **4**A-**4**E with respect to catheters **400***a*-**400***e* are equally applicable to catheters **100** and **200** described above and are not in any way meant to be limiting.

FIG. **4**A illustrates a front view of a catheter **400***a*, in accordance with some embodiments. Catheter **400***a* includes a plurality of shock wave emitters **406***a*-**409***a* arrayed asymmetrically about longitudinal axis **481** of catheter **400***a*. An outer surface of each of shock wave emitters **406***a*-**409***a* is flush with an outer circumferential surface a catheter body **401** of the catheter **400**. FIG. **4**B illustrates a front view of a catheter **400***b*, in accordance with some embodiments. Catheter **400***b* similarly includes a plurality of shock wave emitters **406***b*-**408***b* also arrayed asymmetrically about longitudinal axis **481** of catheter **400***b*. Shock wave emitter **406***b* is positioned such that its outer circumferential surface is positioned externally relative an outer circumferential surface of the catheter body **401***b*, shock wave emitter **407***b* is positioned such that its outer circumferential surface is positioned inset relative an outer circumferential surface of the catheter body **401***b*, and shock wave emitter **408***b* is positioned such that its outer circumferential surface is flush with an outer circumferential surface of the catheter body **401***b*.

FIGS. **4**C-**4**E illustrate a side view of catheters **400***c*-**400***e*, respectively, in accordance with some embodiments. Catheter **400***c* depicted in FIG. **4**C includes a plurality of shock wave emitters **406***c*-**408***c* at a distal end **404** of the catheter body **401**. Shock wave emitters **406***c*-**408***c* are positioned such that a distal most surface of each of shock wave emitters **406***c*-**408***c* is positioned forward of a distal most surface of the distal end **404** of the catheter body **201**. All of shock wave emitters **406***c*-**408***c* are further positioned at the same distal location relative to the distal most surface of the catheter body **401**. Shock wave emitters **406***c*-**408***c* may be arrayed symmetrically or asymmetrically about the longitudinal axis **481** of the catheter body **401**. Catheter **400***d* of FIG. **4**D includes a plurality of shock wave emitters **406***d*-**408***d* at the distal end of catheter body **401**. Shock wave emitters **406***d*-**408***d* are positioned such that a distal most surface of each of shock wave emitters **406***d* and **408***d* is forward of the distal most surface of catheter body **401** and shock wave emitter **407***d* is positioned such that a distal most surface of shock wave emitter **407***d* is recessed from a distal most surface of the catheter body **401**.

FIG. **4**E illustrates a side view of catheter **400***e*, which includes a plurality of shock wave emitters **406***e*-**408***e* at a distal end **404** of the catheter body **401**. Catheter **400***e* further includes an enclosure **490** positioned to cover the plurality of shock wave emitters **406***e*-**408***e*. The enclosure **490** may include any of the characteristics described above with reference to enclosure **190**. Accordingly, in some embodiments, the enclosure **490** is a cap or an angioplasty balloon. In some embodiments, the enclosure **490** can be filled or inflated by pumping a conductive fluid, such as saline and/or contrast agent into the interior volume of the balloon. The enclosure **490** may be inflated before or after applying shock waves to a treatment region. For example, in some embodiments, after forward directed shock waves are initiated using the shock wave emitters **406***e*-**408***e* at the distal end **404** of the catheter **400***e* to break apart an occlusion, the catheter **400***e* is advanced further into a patient's vessel, and the enclosure **490** is inflated in the region of the occlusion to further treat the region. The conductive fluid may aid in cooling the device and dissipating heat generated during the formation of vapor bubbles that result from shock wave generation. The enclosure may also shield the shock wave emitters from direct contact with vessel walls and/or direct contact with an occlusion inside a body lumen. In some embodiments the enclosure **490** is formed of a thin, acoustically transparent material, such as polyethylene or nylon, which can provide for efficient fluid-to-tissue transmission and effective coupling of the pressure pulse from shock wave emitter to the occlusion or other calcification. While the enclosure is only illustrated with respect to the exemplary catheter shown in FIG. **4**E, it should be understood that a similar enclosure could optionally be provided on any of catheters **4**A-**4**D, or any of the other catheters described herein. Alternatively, any of the embodiments described herein may be configured as an open system (i.e., there may be no enclosure surrounding the shock wave emitters provided on any of the catheters described herein).

As described above, the exemplary catheters described herein may be connected to a voltage source and fluid source. The voltage source may be connected to one or more of the wires used to generate shock waves at the plurality of shock wave emitters of the catheters described herein, and the fluid source may be used to fill portions of the catheter, such as enclosure **190** and **490**, with a conductive fluid, as described above.

FIG. **5** illustrates an exemplary system **500** comprising a catheter **520** connected to a shock wave energy generator **530**. In some embodiments, catheter **520** may include any combination of the features described above with reference to FIGS. **1**A-**4**E above, and/or FIGS. **7**A-**11**C below. Accordingly, in some embodiments, catheter **520** includes a plurality of shock wave emitters **506**-**508** at a distal end **524** of catheter **520**. In some embodiments, the plurality of shock wave emitters **506**-**508** include laser emitters, and a plurality of optical fibers extend along the length of the catheter **520** from the emitters **506**-**508** to the shock wave energy generator **530**. In some embodiments, catheter **520** is connected to one or both of the shock wave energy generator **530** and/or fluid source **540** at proximal end **522** of the catheter **520**.

In some embodiments, shock wave energy generator **530** is a portable and/or rechargeable voltage source. In some embodiments shock wave energy generator **530** is a laser pulse generator. In some embodiments, shock wave energy generator **530** is configured to deliver high voltage pulses to a shock wave emitter of a plurality of shock wave emitters of catheter **520**, wherein the high voltage pulses are between 3 kV and 20 kV, including 3 kV and 20 kV. In some embodiments, the high voltage pulses are between 10 kV and 20 kV, including 10 kV and 20 kV. In some embodiments, the high voltage pulses are between 15 kV and 20 kV, including 15 kV and 20 kV. In some embodiments, the high voltage pulses are greater than 20 kV. The high voltage pulses may be at least 1 kV, at least 2 kV, at least 3 kV, at least 4 kV, at least 5 kV, at least 6 kV, at least 7 kV, at least 8 kV, at least 9 kV, at least 10 kV, at least 11 kV, at least 12 kV, at least 13 kV, at least 14 kV, at least 15 kV, at least 16 kV, at least 17 kV, at least 18 kV, at least 19 kV, and/or at least 20 kV. The high voltage pulses may be no more than 20 kV, no more than 19 kV, no more than 18 kV, no more than 17 kV, no more than 16 kV, no more than 15 kV, no more than 14 kV, no more than 13 kV, no more than 12 kV, no more than 11 kV, no more than 10 kV, no more than 9 kV, no more than 8 kV, no more than 7 kV, no more than 6 kV, no more than 5 kV, no more than 4 kV, no more than 3 kV, no more than 2 kV, and/or no more than 1 kV.

In some embodiments, shock wave energy generator **530** is configured to deliver the voltage pulses at a rate of between 1 Hz and 100 Hz, including 1 Hz and 100 Hz. In some embodiments, shock wave energy generator **530** is configured to deliver the voltage pulses at a rate of a rate of between 1 Hz and 50 Hz, including 1 Hz and 50 Hz. Shock wave energy generator **530** may be configured to deliver the voltage pulses at a rate of a rate of up to 100 Hz, up to 90 Hz, up to 80 Hz, up to 70 Hz, up to 60 Hz, up to 50 Hz, up to 40 Hz, up to 30 Hz, up to 20 Hz, and/or up to 10 Hz. Shock wave energy generator **530** may be configured to deliver the voltage pulses at a rate of at least 10 Hz, at least 20 Hz, at least 30 Hz, at least 40 Hz, at least 50 Hz, at least 60 Hz, at least 70 Hz, at least 80 Hz, at least 90 Hz, and/or at least 100 Hz. In some embodiments, shock wave energy generator **530** is configured to deliver the voltage pulses at a rate of a rate of up to 20 Hz, including at a rate of 20 Hz. In some embodiments, shock wave energy generator **530** can be configured to apply an alternating current to the electrodes of the shock wave emitters to induce a change in the polarity of the electrodes.

In some embodiments, catheter **520** is connected to fluid source **540** using a pump **550**. In some embodiments, the fluid source **540** may contain a conductive fluid such as saline and/or contrast agent that can be injected into catheter **520** using the pump **550**. As described above, the conductive fluid injected into catheter **520** may be used to fill an inflatable enclosure, such as a cap or an angioplasty balloon. The conductive fluid may aid in cooling the device and dissipating heat generated during the formation of vapor bubbles that result from shock wave generation. The enclosure may also shield the shock wave emitters from direct contact with vessel walls and/or direct contact with an occlusion inside a body lumen.

The catheters described herein may be used for treating various calcifications within a human body lumen, according to some embodiments. For instance, the shock waves generated by any of catheters **100**, **200**, **400**a-**400**e, and/or **520** may be used to crack, fragment, or otherwise break up calcifications such as kidney stones and/or occlusions within a blood vessel. FIG. **6** illustrates an exemplary method for emitting shock waves in a body lumen, according to some embodiments.

At block **602**, an exemplary catheter that includes a plurality of shock wave emitters located at a distal end of the catheter body is positioned adjacent to an occlusion in a vessel. The vessel may include blood vessels in a patient's vascular system or ureters in the patient's urinary system. The catheter may be any of catheters **100**, **200**, **400**a-e, or **500** described above, and/or any of catheters **700**a-**700**b, **800**a-**800**c, **900**, **1000**, and/or **1100**a-**1100**c below. The shock wave device is positioned within the vessel such that a distal end of the device faces a first treatment region. The first treatment region may include a chronic total occlusion (CTO), circumferential calcium, a kidney stone, or other obstructions or concretions. Once the distal end of the shock wave device is facing the first treatment region, the method **600** can proceed to block **604**. In some embodiments, the first treatment region may include fibrotic/calcified cardiac tissue surrounding a pacemaker lead, and the catheter may be positioned such that a lumen of the catheter body surrounds the pacemaker lead, for instance as shown in FIG. **9**.

At block **604**, the catheter emits a first plurality of shock waves from the plurality of shock wave emitters in a distal direction so that the shock waves constructively interfere distally of the distal end of the catheter. As described above, the shock waves may be generated by applying a voltage across a plurality of electrode pairs, each separated by a spark gap. A spark is generated as a current flows between the electrodes of the electrode pair across the spark gap, which generates a shock wave. This shock wave generation process may occur simultaneously across a plurality of shock wave emitters connected in series and/or shock waves may be generated at one or more shock wave emitters configured to generate shock waves independently of the other emitters. Due to the positioning of the shock wave emitters, the plurality of shock waves propagate in a substantially forward direction out of the catheter to constructively interfere with one another and impinge on the occlusion or calcium in the first treatment area.

At block **606**, the catheter is optionally advanced further into the blood vessel, for instance, if the first plurality of shock waves did not successfully break up the entire occlusion. At block **608**, the catheter optionally emits a second plurality of shock waves from the plurality of shock wave emitters so that the shock waves constructively interfere at a location that is distal of the first location. Blocks **606** and **608** may be iteratively repeated until the occlusion has been successfully treated. At block **610**, the catheter is optionally used to remove a pacemaker lead from a cardiac tissue. As noted above, the pacemaker lead may be inserted into a lumen of the catheter body. The shock waves emitted by the plurality of shock wave emitters may fragment/break up fibrotic/calcified tissue around the pacemaker lead, allowing the pacemaker lead to be removed by withdrawing the catheter away from the treatment area. It should be understood that the aforementioned steps do not need be performed in the order presented, and some steps may be omitted altogether. For instance, the pacemaker lead may be removed immediately following the initial plurality of shock waves generated at step **604** and a second plurality of shock waves may not be emitted from the catheter.

As described throughout, the catheters described herein may include an enclosure that provides a protective barrier between the shock wave emitters and soft tissue within the body by mitigating thermal injury to soft tissue and reducing cavitation stresses by limiting expansion of the vapor bubble to the interior of the cap. FIG. **7**A illustrates the impact of an enclosure **790** disposed at a distal end of a catheter body **701**, encasing shock wave emitters **706**a-**708**a impacts the propagation of cavitation bubbles distally of the distal end of the catheter body **701**. As shown, the vapor bubbles hit the wall of enclosure **790** before reaching their maximum potential size, thus inducing collapse preventing potential soft tissue injury that can be caused by tensile stresses resulting from the collapse of the bubbles outside of the enclosure **790**. In the embodiment illustrated in FIG. **7**A, the enclosure **790** is approximately semi-cylindrical or semi-spherical in shape. FIG. **7**B illustrates that various differently shaped caps could be used in place of the semi-cylindrical or semi-spherical shape, for instance enclosure with either greater or less elongation distally of the distal end of the catheter body **701**b. In some embodiments, an enclosure having greater elongation distally of distal end of the catheter body **701**b (i.e., having a narrower parabolic shape) may result in shock waves emitted from the shock wave emitters and catheter having a correspondingly narrower parabolic shape. In some embodiments, an enclosure having less elongation distally of distal end of the catheter body **701**b (i.e., having a wider parabolic shape) may result in shock waves emitted from the shock wave emitters having a correspondingly wider parabolic shape.

In some embodiments, no enclosure is provided on the catheters described herein. In such embodiments, it may be desirable to exercise greater control over the direction of shock wave and/or cavitation bubble propagation to direct cavitation bubbles and shock waves away from soft tissue and toward target treatment areas. FIGS. **4**A-**4**D described above illustrated various options for arranging shock wave emitters at the distal end of a catheter. FIGS. **8**A-**8**C described below further illustrate how both the arrangement of emitters and the shape of the catheter body itself can impact the manner in which shock waves propagate distally of the distal end of the catheter body. In some embodiments, positioning shock wave emitters such that their distal most surface is recessed from a distal most surface of the catheter body (i.e., into the catheter body may cause the recessed shock wave emitter to generate cavitation bubbles and corresponding shock waves that propagate at an angle diverging from the longitudinal axis of the shock wave emitter and/or the longitudinal axis of the catheter body. In some embodiments, configuring the distal most surface of the catheter body such that it is slanted (i.e., not perpendicular to the longitudinal axis of the catheter body) may cause cavitation bubbles and corresponding shock waves generated by the shock wave emitters to propagate at an angle diverging from the longitudinal axis of the catheter body.

FIG. **8**A illustrates a side view of catheter **800**a including a plurality of shock wave emitters **806**a-**808**a at a distal end of a catheter body **801**. Each of shock wave emitters **806**a-**808**a are positioned such that a distal most surface of each shock wave emitter is respectively positioned flush with a flat (i.e., perpendicular to a longitudinal axis **812**a of the catheter body **801**) distal most surface **804**a of catheter body **801**. In some embodiments, the distal most ends of each emitter are coplanar with each other and/or a distal most edge surface of the catheter body. In some embodiments, one or more shock wave emitters have distal most ends that extend no more than 0.5 mm longitudinally from a flat distal most surface of the catheter body. In some embodiments, one or more shock wave emitters have distal most ends that extend no more than 0.1 mm, no more than 0.2 mm, no more than 0.3 mm, no more than 0.4 mm, no more than 0.5 mm, no more than 0.6 mm, no more than 0.7 mm, no more than 0.8 mm, no more than 0.9 mm, or no more than 1.0 mm longitudinally from a flat distal most surface of the catheter body. In some embodiments, one or more shock wave emitters have distal most ends that extend at least 0.1 mm, at least 0.2 mm, at least 0.3 mm, at least 0.4 mm, at least 0.5 mm, at least 0.6 mm, at least 0.7 mm, at least 0.8 mm, at least 0.9 mm, or at least 1.0 mm. In some embodiments, one or more shock wave emitters have distal most ends that extend between 0.01 mm and 1.0 mm longitudinally from a flat distal most surface of the catheter body. In some embodiments, positioning a distal most surface of the shock wave emitters flush with a flat distal most surface **804***a* of the catheter body may result in cavitation bubbles that propagate distally of the distal end of the catheter body **801** along each of the respective longitudinal axes (**809***a*, **810***a*, and **811***a*) of the plurality of shock wave emitters. The cavitation bubbles may collapse distally of the distal end of catheter body **801**, resulting in corresponding constructively interfering shock waves that propagate forward in a unitary direction, as illustrated. In some embodiments, such a configuration of both the shock wave emitters and the catheter body may result in cavitation bubbles that propagate distally of the distal end of catheter body **801** in a direction parallel with the longitudinal axis **812***a* of the catheter body **801**.

FIG. **8**B illustrates that positioning shock wave emitters **806***b*-**808***b* in a staggered formation relative to a flat distal most surface **804***b* of catheter body **801** may enable directional control of shock waves. In the embodiment of FIG. **8**B, a distal most surface of shock wave emitter **806***b* is positioned flush with a distal most surface **804***b* of catheter body **801**, a distal most surface of shock wave emitter **807***b* is recessed a first distance from the distal most surface **804***b* of catheter body **801**, and a distal most surface of shock wave emitter **808***b* is recessed a second distance, greater than the first distance, from the distal most surface **804***b* of catheter body **801**. As described above with reference to FIG. **8**A, shock wave emitter **806***b* may produce cavitation bubbles that propagate distally of the distal end of the catheter body **801** along the longitudinal axis **809***b* of shock wave emitter **806***b*. In contrast, the positioning of shock wave emitters **807***b* and **808***b* may result in cavitation bubbles that propagate distally of the distal end of the catheter body **801** at respective angles relative to the longitudinal axes of shock wave emitters **807***b* and **808***b*, such that the cavitation bubbles diverge from the longitudinal axes **810***b* and **811***b*, respectively, of shock wave emitters **807***b* and **808***b*. In some embodiments, the recession of the distal most surface of a shock wave emitter further into the catheter body **801** may result in cavitation bubbles that diverge further (i.e., at an angle greater in magnitude) from the longitudinal axis of the respective shock wave emitter. In some embodiments, the shock waves resulting from collapse of the cavitation bubbles may constructively interfere with one another and propagate forward at an angle relative to the longitudinal axis **812***b* of catheter body **801**, wherein the direction of propagation of the constructively interfering shock waves is based on the respective direction of propagation of the cavitation bubbles generated by each of the shock wave emitters.

FIG. **8**C illustrates that the shape of the catheter body may also be configured to enable directional control of shock waves. In the embodiment of FIG. **8**C, a distal most surface **804***c* of catheter body **801** is slanted, rather than perpendicular to a longitudinal axis **812***c* of the catheter body **801** (i.e., configured such that a plane corresponding to the distal most surface **804***c* intersects a plane corresponding to the longitudinal axis **812** of catheter body **801** at an angle greater than or less than 90 degrees). The catheter **800***c* of FIG. **8**C includes a plurality of shock wave emitters **806***c*-**808***c*. The shock wave emitters **806***c*-**808***c* are positioned such that a distal most surface of each shock wave emitter is positioned at a different distal location relative to each of the other shock wave emitters. In the embodiment of FIG. **8**C, a distal most surface of shock wave emitter **806***c* is positioned at a first location relative to a proximal end of catheter body **801**, a distal most surface of shock wave emitter **807***c* is positioned at a second location relative to the distal most surface of shock wave emitter **806***c*, and a distal most surface of shock wave emitter **808***c* is positioned at a third location relative to the distal most surfaces of both shock wave emitters **806***c* and **807***c*. Optionally, the distal most surface of each of the shock wave emitters **806***c*-**808***c* may be positioned flush with the distal most surface **804***c* of catheter body **801**.

The relative positioning of shock wave emitters **806***c*-**808***c* and the slanted distal most surface **804***c* of catheter body **801** may result in cavitation bubbles generated by each of the shock wave emitters that propagate distally of the distal end of the catheter body **801** parallel to one another at an angle diverging from each of their respective longitudinal axes **809***c*, **810***c*, and **811***c*. Thus, the cavitation bubbles generated by each of the shock wave emitters **806***c*-**808***c* propagate parallel to one another at the same angle diverging from the longitudinal axis **812***c* of the catheter body **801**, and accordingly, the shock waves resulting from collapse of the cavitation bubbles may constructively interfere with one another and propagate forward in the same direction as the cavitation bubbles relative to the longitudinal axis **812***c* of catheter body **801**.

As described above, the catheters described herein may be useful for pacemaker lead removal. Cardiac tissue surrounding pacemaker leads inserted into the tissue can become calcified/fibrotic over time, making it difficult to remove the leads. Shock wave treatment can break up the calcified/fibrotic tissue, thus loosening the attachment between the tissue and the pacemaker leads and enabling easier removal of the leads. FIG. **9** illustrates an exemplary catheter **900** including a plurality of shock wave emitters and a lumen **904** for removing a pacemaker lead **906** from cardiac tissue **908**. The catheter **900** may be positioned within a vessel such that pacemaker lead **906** is inserted into lumen **904**. Pacemaker lead **906** may thus act as a guidewire for catheter **900**, enabling a user to advance the catheter to a point of attachment between the pacemaker lead and cardiac tissue (e.g., heart muscle). When the catheter is positioned such that the shock wave emitters are adjacent to and facing the calcified/fibrotic tissue **910**, a plurality of shock waves can be generated (as described herein) to break up the calcified/fibrotic tissue **910** and loosen the pacemaker lead. If the first plurality of shock waves do not free the pacemaker lead from the tissue, the catheter **900** can optionally be advanced further toward the calcified/fibrotic tissue **910** and/or a second plurality of shock waves can be generated. This process can be repeated as needed until the attachment between the pacemaker lead and cardiac tissue is sufficiently loosened such that the pacemaker lead can be removed either by removing the catheter along with the pacemaker lead, or by removing the catheter and inserting a separate tool for pacemaker lead removal.

As described throughout, a plurality of shock wave emitters positioned at the distal end of the catheters described herein may be electrically connected in series such that applying a voltage pulse across two electrodes causes each of the shock wave emitters to generate a shock wave. In some embodiments, one or more shock wave emitters may be configured to fire independently. FIG. **10** illustrates an exemplary catheter **1000** including shock wave emitters **1006** and **1008** configured to fire independently of one another and of the other shock wave emitters **1012**, **1014**, **1016**, and **1018** included on catheter **1000**. Shock wave emitter **1006** includes an electrode pair formed by an exposed end of an insulated wire **1022** and a conductive band **1070**. The insulated wire **1022** extends into the conductive band **1070** from a proximal end of the conductive band toward a distal end of the conductive band **1070** and is positioned such that the exposed end of the wire **1022** is separated by a spark gap from conductive band **1070**. The insulated wire **1022** may extend from the proximal end of the conductive band **1070** along the length of the catheter body **1001** of catheter **1000** to connect to a first terminal of a voltage source. A second insulated wire **1020** may extend into the conductive band **1070** from the proximal end of the band and connect to (e.g., may be soldered, crimped, tapes, clamped, or otherwise connected to) a surface of the conductive band **1070**. The second insulated wire may extend from the proximal end of the conductive band **1070** along the length of the catheter body to connect to a second terminal of the voltage source. The insulated wires **1020** and **1022**, respectively, may be connected to a negative and positive terminal of the voltage source. Accordingly, when a voltage is applied across wire **1020** and **1022**, a shock wave is generated at shock wave emitter **1006**, but the voltage applied across wire **1020** and **1022** does not result in shock waves at any of the other shock wave emitters provided on catheter **1000**.

Similarly, shock wave emitter **1008** includes an electrode pair formed by an exposed end of an insulated wire **1026** and a conductive band **1072**. The insulated wire **1026** extends into the conductive band **1072** from a proximal end of the conductive band toward a distal end of the conductive band **1072** and is positioned such that the exposed end of the wire **1026** is separated by a spark gap from conductive band **1072**. The insulated wire **1026** may extend from the proximal end of the conductive band **1072** along the length of the catheter body **1001** of catheter **1000** to connect to a first terminal of a voltage source. A second insulated wire **1024** may extend into the conductive band **1072** from the proximal end of the band and connect to (e.g., may be soldered, crimped, tapes, clamped, or otherwise connected to) a surface of the conductive band **1072**. The second insulated wire may extend from the proximal end of the conductive band **1072** along the length of the catheter body to connect to a second terminal of the voltage source. The insulated wires **1026** and **1024**, respectively, may be connected to a negative and positive terminal of the voltage source. Accordingly, when a voltage is applied across wire **1024** and **1026**, a shock wave is generated at shock wave emitter **1008**, but the voltage applied across wire **1024** and **1026** does not result in shock waves at any of the other shock wave emitters provided on catheter **1000**. In contrast, the plurality of shock wave emitters **1012**, **1014**, **1016**, and **1018** are electrically connected in series and thus fire simultaneously with one another, for instance, in the same manner described above with reference to the shock wave emitters of catheter **200** illustrated in FIG. **3**A.

Constructive interference of the shock waves generated by the shock wave emitters described herein involves the combination/overlap of shock waves produced by each of the respective emitters distally of the distal end of the catheter. The combination of the individual shock waves results in a combined shock wave that has a higher pressure than the shock waves emitted by each individual emitter, which enables the combined shock wave to treat denser and more rigid calcified lesions in the body. FIGS. **11**A-**11**C illustrate constructive interference of shock waves generated by a plurality of shock wave emitters, and how the shape of the catheter body and/or the positioning of the shock wave emitters may affect the direction of shock wave propagation and resulting constructive interference according to some embodiments. In some embodiments, constructive interference of pressure waves may be controlled by syncing or offsetting the timing of delivery of energy pulses to individual emitters.

FIG. **11**A illustrates a side view of a catheter **1100*a*** including a plurality of shock wave emitters **1106*a*-1108*a*** at a distal end of a catheter body **1101**. Each of shock wave emitters **1106*a*-1108*a*** are positioned such that a distal most surface of each shock wave emitter is respectively positioned flush with a flat (i.e., perpendicular to a longitudinal axis of the catheter body **1101**) distal most surface **1104*a*** of catheter body **1101**. In some embodiments, positioning a distal most surface of the shock wave emitters flush with a flat distal most surface of the catheter body **1101** may result in shock waves from each of the respective emitters that propagate distally of the distal end of the catheter body **1101** along each of the respective longitudinal axes **1109*a***, **1110*a***, and **111*a***, of the plurality of shock wave emitters. The shock waves generated by each of the respective shock wave emitters **1106*a*-1108*a*** may propagate forward in a unitary direction, as illustrated, and constructively interfere with one another distally of the distal end of the catheter body **1101**. More specifically, each shock wave emitter **1106*a*-1108*a*** may generate a respective shock wave. As each of the shock waves propagate forward distally of the distal end of catheter body **1101**, the shock waves expand and begin to overlap with the shock waves generated by each of the other emitters. The overlapping shock waves constructively interfere with one another to form a combined shock wave that continues to propagate forward distally of the distal end of the catheter body **1101**. In some embodiments, the combined shock wave propagates distally of the distal end of the catheter body **1101** in the direction of the longitudinal axis **1112*a*** of the catheter body **1101**.

FIG. **11**B illustrates a side view of a catheter **1100*b*** including a plurality of shock wave emitters **1106*b*-1108*b*** at a distal end of a catheter body **1101**. FIG. **11**B illustrates that the shape of the catheter body may be configured to enable directional control of shock waves. In the embodiment of FIG. **11**B, a distal most surface **1104*b*** of catheter body **1101** is slanted, rather than perpendicular to a longitudinal axis **1112*b*** of the catheter body **1101** (i.e., configured such that a plane corresponding to the distal most surface **1104*b*** of the catheter body **1101** intersects a plane corresponding to the longitudinal axis **1112*b*** of catheter body **1101** at an angle greater than or less than 90 degrees). The catheter **1100**B of FIG. **11**B includes a plurality of shock wave emitters **1106*b*-1108*b***. The shock wave emitters **1106*b*-1108*b*** are positioned such that a distal most surface of each shock wave emitter is positioned at a different distal location relative to each of the other shock wave emitters. In the embodiment of FIG. **11**B, a distal most surface of shock wave emitter **1106***b* is positioned at a first location relative to a proximal end (left side of the catheter **1100** from the view of FIG. **11**B) of catheter body **1101**, a distal most surface of shock wave emitter **1107***b* is positioned at a second location relative to the distal most surface of shock wave emitter **1106***b*, and a distal most surface of shock wave emitter **1108***b* is positioned at a third location relative to the distal most surfaces of both shock wave emitters **1106***b* and **1107***b*. Optionally, the distal most surface of each of the shock wave emitters **1106***b*-**1108***b* may be positioned flush with the distal most surface **1104***b* of catheter body **1101**.

The relative positioning of shock wave emitters **1106***b*-**1108***b* and the slanted distal most surface of catheter body **1101** may result in shock waves generated by each of the shock wave emitters that propagate distally of the distal end of the catheter body **1101** parallel to one another at an angle diverging from each of their respective longitudinal axes **1109***b*, **1110***b*, and **1111***b*. Thus, the shock waves generated by each of the shock wave emitters **1106***b*-**1108***b* propagate parallel to one another at the same angle diverging from the longitudinal axis **1112***b* of the catheter body **1101**, and accordingly, the shock waves may constructively interfere with one another and propagate forward in the same direction as the shock waves generated by each of the respective shock wave emitters relative to the longitudinal axis **1112***b* of catheter body **1101**.

FIG. **11**C illustrates that the positioning shock wave emitters **1106***c*-**1108***c* relative to a distal most surface **1104***c* of the catheter body **1101** (for instance, in a staggered formation relative to a flat distal most surface of catheter body **1101**) may enable directional control of shock waves. In the embodiment of FIG. **11**C, a distal most surface of shock wave emitter **1106***c* is positioned flush with a distal most surface **1104***c* of catheter body **1101**, a distal most surface of shock wave emitter **1107***c* is recessed a first distance from the distal most surface **1104***c* of catheter body **1101**, and a distal most surface of shock wave emitter **1108***c* is recessed a second distance, greater than the first distance, from the distal most surface **1104***c* of catheter body **1101**.

Shock wave emitter **1106***c* may produce shock waves that propagate distally of the distal end of the catheter body **1101** along the longitudinal axis **1109***c* of shock wave emitter **1106***c*. In contrast, the positioning of shock wave emitters **1107***c* and **1108***c* may result in shock waves that propagate distally of the distal end of the catheter body **1101** at respective angles relative to the longitudinal axes **1110***c* and **1111***c* of shock wave emitters **1107***c* and **1108***c*, such that the shock waves diverge from the longitudinal axes of shock wave emitters **1107***c* and **1108***c*. In some embodiments, the recession of the distal most surface of a shock wave emitter further into the catheter body **1101** may result in shock waves that diverge further (i.e., at an angle greater in magnitude) from the longitudinal axis of the respective shock wave emitter. In some embodiments, the shock waves constructively interfere with one another as illustrated and propagate forward at an angle relative to the longitudinal axis **1112***c* of catheter body **1101**, wherein the direction of propagation of the constructively interfering shock waves is based on the respective direction of propagation of the individual shock waves generated by each of the shock wave emitters.

FIGS. **12**A-**12**C illustrate exemplary catheters that include a plurality of shock wave emitters according to some embodiments of the disclosure. In some embodiments, the catheter is configured to emit shock waves in a primarily radial direction (i.e., transverse to a longitudinal axis of the catheter) from a plurality of shock wave emitters. Similar to embodiments described above, the respective shock wave emitters may include an electrode pair defined by ends of a pair of conductive wires separated by a spark gap. However, other configurations (e.g., an electrode pair defined by a wire end and a conductive sheath, emitter band, or other conductive material, such as the configuration illustrated in FIGS. **3**A, **3**B, and **10** above) are also possible. FIG. **12**A illustrates an isometric view of a catheter **1200** that includes a shock wave emitter **1215** that includes pair of electrodes **1202** and **1204** separated by a spark gap. Electrodes **1202** and **1204** are defined by respective ends of first and second conductive wires. The first and second conductive wires **1262** and **1264** may extend along at least a portion of the length of the catheter body **1201** to connect to a positive and negative terminal of a voltage source. When a voltage is applied across the electrodes **1202** and **1204**, a shock wave may be generated, as described throughout.

In some embodiments, the pair of electrodes **1202** and **1204** that form shock wave emitter **1215** are positioned adjacent to a surface **1220**. In some embodiments, electrodes **1202** and **1204** are positioned within a cavity **1250** of surface **1220**. In some embodiments, at least one of the cavity **1250** (e.g., a concave, rectangular, or other carve-out of surface **1220**) and surface **1220** are configured to cause shock waves that are generated when a voltage (pulse) is applied across the electrodes **1202** and **1204** to propagate outward from the surface **1220** (e.g., radially from the catheter **1200** with respect to longitudinal axis **1281**). In other words, the surface **1220** and/or cavity **1250** may be configured to reflect shock waves generated by the emitter. Accordingly, at least one of the cavity **1250** and surface **1220** may be formed at least in part of a material that at least partially reflects shock waves. The generated shockwaves thus may be at least partially reflected from the surface **1220** and/or cavity **1250**, focusing the shockwaves in a radially outward direction from the surface and cavity. In some embodiments, at least a portion of surface **1220** is formed of one of aluminum, nitinol, stainless steel, or an alloy thereof.

In some embodiments, the surface **1220** may be planar, curved, or any other geometry configured to cause shock waves generated by shock wave emitter **1215** to propagate radially outward from the surface **1220**. The geometry of the surface may impact the form and manner of propagation of the shock waves. For instance, an angle of tilt of the surface **1220** may enable directional control over shock waves, thus allowing for more targeted treatment. In some embodiments, the magnitude of curvature of the surface **1220** may impact the form of the emitted shock waves. For example, a higher magnitude of curvature of surface **1220** may result in shock waves having a relatively smaller radius, and a lower magnitude of curvature of surface **1220** may result in shock waves having a relatively larger radius.

According to some embodiments, and as illustrated in FIG. **12**A, the catheter body **1201** includes a first portion **1280** and a second portion **1290** separated by a chamber **1285**. The second portion **1290** may be distal of the shock wave emitter **1215** and may form a distal end of catheter **1200**. In some embodiments, at least one of the first portion **1280** and the second portion **1290** of catheter **1200** are formed at least in part of a material that does not readily transmit emitted shock waves. In some embodiments, at least a portion of at least one of the first portion **1280** and the second portion **1290** of catheter **1200** are formed of one of aluminum, nitinol, stainless steel, or an alloy thereof.

In some embodiments, the plurality of shock wave emitters are positioned at least partially within the chamber **1285**. The first portion **1280** of catheter body **1201** may include a surface **1270** positioned at a proximal end of chamber **1285**, adjacent to the chamber **1285**. The second portion **1290** of catheter body **1201** may include a surface **1240** positioned at a distal end of chamber **1285**, adjacent to the chamber **1285**. The surfaces **1270** and **1240**, similarly to the first portion of catheter body **1280** and second portion of the catheter body **1290**, may be formed at least in part of a material that does not readily transmit emitted shock waves (e.g., aluminum, nitinol, stainless steel, or an alloy thereof). Accordingly, the surfaces **1270** and **1240** may further impact the shape and manner of propagation of shock waves emitted from shock wave emitter **1215**, for instance, by reflecting shock waves at least partially in a longitudinal direction upon contact with surfaces **1270** and **1240** and/or constraining the propagation of the shock waves along the longitudinal axis **1281** of catheter **1200** within the chamber **1285**.

In some embodiments, an enclosure **1260** (e.g., an acoustically transparent window) at least partially circumscribes the plurality of shock wave emitters and defines the chamber **1285**. In some embodiments, the enclosure **1260** defines an outer diameter of the chamber **1285**. The enclosure **1260** facilitates transmission of shock waves from the inner chamber to a target treatment site. For instance, similarly to the enclosure **190** described above, the enclosure/acoustically transparent window **1260** can be filled or inflated by pumping a conductive fluid, such as saline and/or contrast agent into its interior volume. As described above, without an enclosure, cavitation bubbles formed during shock wave generation may exert tensile forces on a target treatment area when they collapse upon impact with the treatment area. However, with enclosure **1260** enclosing the radially firing shock wave emitters, the cavitation bubbles resulting from shock wave generation are separated from the target treatment area. Accordingly, the enclosure can protect soft tissue from potential damage resulting from the cavitation bubbles.

In some embodiments the enclosure **1260** is formed of a thin, acoustically transparent material, such as polyethylene or nylon, which can provide for efficient fluid-to-tissue transmission and effective coupling of the pressure pulse from shock wave emitter to the occlusion or other calcification. In some embodiments, the catheter **1200** may be connected to a fluid source **540** and fluid pump **550** (e.g., as shown in FIG. **5**). The fluid pump **550** may fill the interior volume of the enclosure **1260** with fluid to a certain pressure. The conductive fluid may be circulated within the interior volume of the enclosure (e.g., chamber **1285**) via a lumen of the catheter **1200**, for instance lumen **1252** described below with reference to FIG. **12**C, by injecting the fluid into the enclosure **1260** by a fluid pump **550** shown in FIG. **5** and drawing it into a fluid return line (e.g., lumen **1254** described below with reference to FIG. **12**C).

In some embodiments, the enclosure **1260** may not be included on catheter **1200**, for instance, as shown in FIG. **12**B. In some embodiments, if the enclosure **1260** is omitted from catheter **1200**, the cavitation stresses resulting from cavitation bubble collapse at the target treatment areas may exert additional force on the treatment area to break up dense calcifications.

FIG. **12**C illustrates a detailed isometric cutaway view of an example of catheter **1200** that depicts a plurality of shock wave emitters and exemplary interior features of the example of the catheter body **101**. The catheter **1200** may include a plurality of shock wave emitters **1215**, **1217**, and **1219**. Similar to shock wave emitter **1215**, shock wave emitter **1217** may include a first electrode **1206** and second electrode **1208** separated from one another by a second spark gap, and shock wave emitter **1219** may include a first electrode **1210** and second electrode **1212** separated from one another by a third spark gap. In some embodiments, the respective electrodes of shock wave emitters **1215**, **1217**, and **1219** are defined by the exposed distal end of a conductive wire; although, as described throughout, one of the electrodes at each emitter may instead be formed of a conductive sheath, band, or other conductive material. At least two of the wires may respectively extend along the length of the catheter body **1201** to connect to positive and negative terminals of a voltage source. For instance, each of the wires may extend along the length of the catheter body to connect to a respective positive and negative terminal of a voltage source. Accordingly, each of the respective shock wave emitters **1215-1217** may be configured such that it can be fired (e.g., generate shock waves) independently of each of the other shock wave emitters provided on the catheter **1200**. In some embodiments, a wire at a first emitter may extend along the length of the catheter to connect to a positive terminal, and a wire at a second emitter may extend along the length of the catheter to connect to a negative terminal, and the remaining wires may connect each of the respective shock wave emitters **1215**, **1217**, and **1219** in series such that when a voltage is applied across the two wires connected to the voltage source, a shock wave is generated by each of the shock wave emitters **1215**, **1217**, and **1219**, for instance as described above with reference to catheter **100** illustrated in FIG. **1**B. In some embodiments, at least two of the shock wave emitters (e.g., emitters **1215** and **1217**) may be connected in series, and the remaining emitter (e.g., **1219**) may be configured to fire independently of the two connected in series. In some embodiments, the wires may extend into a respective lumen of the catheter body that extends from the chamber **1285** to a proximal end of the catheter body **1201**.

Shock wave emitters **1215**, **1217**, and **1219** may be evenly spaced (positioned at increments of about 120 degrees/circumferentially uniformly distributed about the longitudinal axis) about the longitudinal axis **1281**. Although, in some embodiments, as discussed throughout, the **1215**, **1217**, and **1219** may instead be unevenly spaced (e.g., **1215** and **1217** may be separated by 100 degrees and **1217** and **1219** may be separated by 260 degrees, and so on). For instance, more shock wave emitters may be positioned on one side (e.g., along a dorsal and ventral side of the catheter) than the opposite side. Further, while the catheter **1200** is illustrated as including three shock wave emitters, it should be understood that additional emitters could be included without deviating from the scope of the disclosure.

As illustrated in FIG. **12**C, in one or more embodiments, shock wave emitters are generally coplanar with a virtual plane **1203** that is perpendicular to the longitudinal axis **1281** of the catheter. In some embodiments, the shock wave emitters may be positioned at staggered locations along the longitudinal axis, as described throughout. For instance, a first emitter may be positioned distally of a second emitter, a third emitter may be positioned distally of one or both of the first and second emitters, and so on. In various embodiments, shock waves generated by emitters **1215**, **1217**, and **1219** and emitted from the radially-emitting catheter **1200** may constructively interfere with one another as they propagate outwardly from the catheter **1200**, for instance, as described in U.S. Pat. No. 11,779,363, which is incorporated herein by reference in its entirety.

In some embodiments, as illustrated in FIG. **12**C, catheter **1200** includes a plurality of lumens, such as lumens **1250**, **1252**, and **1256** formed into the catheter body **1201**. The lumens **1250**, **1252**, and **1256** may extend along the length of the catheter body **1201** to serve a variety of purposes. For instance, in some embodiments, lumen **1252** may be configured to receive and/or fluidly connect to a fluid supply line and lumen **1254** may be configured to receive and/or fluidly connect to a fluid return line. The lumens may extend to the chamber **1285** such that the chamber can be filled with a conductive fluid to facilitate shock wave transmission. In some embodiments, lumen **1256** may be configured to receive a guide wire, similarly to central lumen **150** described above.

FIGS. **13**A and **13**B illustrate a catheter **1300** that includes both forward firing and radially firing shock wave emitters, according to some embodiments. Catheter **1300** may include any combination of the features described above with reference to FIGS. **1**A-**12**C. In some embodiments, catheter **1300** includes a first plurality of shock wave emitters **1302** disposed at a distal end of the catheter body. In some embodiments, each shock wave emitter of the plurality of shock wave emitters disposed at a distal end of the catheter body is configured to generate a shock wave that propagates distally of the catheter body. The first plurality of shock wave emitters may include any of the features described above with reference to the shock wave emitters illustrated in FIGS. **1**A-**11**C.

In some embodiments, catheter **1300** also includes a second plurality of shock wave emitters **1304** positioned proximally of the distal end of the catheter body. In some embodiments, each of the second plurality of shock wave emitters are respectively configured to generate shock waves that propagate radially from the catheter body. The second plurality of shock wave emitters **1304** may include any of the features described above with reference to the shock wave emitters illustrated in FIGS. **12**A-**12**C.

In some embodiments, the first plurality of shock wave emitters is configured to generate a first plurality of shock waves **1302** independently of the second plurality of shock wave emitters **1304**. In some embodiments, the first plurality of shock wave emitters **1302** may be configured to generate a plurality of shock waves simultaneously with the second plurality of shock wave emitters **1304**. In some embodiments, any respective shock wave emitter of both the first plurality of shock wave emitters and the second plurality of shock wave emitters may be configured to generate shock waves independently of any of the other shock wave emitters. Accordingly, a user of catheter **1300** may choose between forward firing, side firing, and/or simultaneous forward and side firing. Further, the user may choose between firing any one of the shock wave emitters individually, firing the first plurality of emitters in unison, firing the second plurality of emitters in unison, firing all emitters in unison, or any combination thereof.

As discussed above, catheters according to the principles described herein can be used to facilitate removal of pacemaker leads, which often become encased in fibrotic tissue. FIG. **14** illustrates a method **1400** for using a shock wave catheter for removing a pacemaker lead. Method **1400** may be performed by any of the catheters described herein.

At step **1402**, a proximal end of a pacemaker lead is inserted into a central lumen of a shock wave catheter, such as through central lumen **150** of catheter **100**, outside of the body. At step **1404**, the catheter is advanced into and through vasculature of the body to the target treatment site in the heart, using the pacemaker lead as a guide. The target treatment site may be the location where the pacemaker lead anchors into the heart wall.

At step **1406**, one or more shock waves are generated by one or more shock wave emitters of the catheter. Step **1406** may include using one or more forward firing shock wave emitters to break up fibrotic tissue located in front of a distal end of the catheter. At least a portion of the fibrotic tissue encasing the pacemaker lead may be positioned adjacent to the shock wave emitters, such as illustrated in FIG. **9**, and the distally propagating shock waves may impinge on the fibrotic tissue, breaking it up.

In some embodiments, the catheter may also include a one or more shock wave emitters positioned proximally of the distal end of the catheter body and configured to generate shock waves that propagate radially from the catheter body, for instance, as described above with reference to FIG. **13**. Accordingly, the method **1400** may include generating a plurality of shock waves using the one or more shock wave emitters configured to generate shock waves that propagate radially from the catheter body to at least partially break up a calcified region of vasculature leading to the target site.

It may be desirable to arrange the shock wave emitters and corresponding electrodes such that when the shock wave emitters of a catheter are simultaneously driven, a single cavitation bubble is formed that propagates distally of the distal end of the shock wave emitters (i.e., in contrast to a plurality of relatively smaller cavitation bubbles that may constructively interfere distally of the distal end). This may be accomplished, for instance, as described below with reference to FIG. **15**, by configuring the shock wave emitters such that each of the shock wave emitters shares a common electrode. The common electrode may be positioned at a location between each of the other electrodes (e.g., the common electrode may be positioned at a center of a distal end of the catheter and an electrode of each respective shock wave emitter may be positioned at an equal distance from the common electrode about a circumference of the distal end of the catheter).

FIG. **15** illustrates a front view of an exemplary catheter **1500** for generating shock waves according to some embodiments. Catheter **1500** may include a plurality of shock wave emitters **1501**, **1503**, and **1505**. Each of the shock wave emitters **1501**, **1503**, and **1505** may include a respective electrode **1502**, **1504**, and **1506** connected to a respective positive terminal (e.g., supply terminal) of a voltage source (e.g., shock wave energy generator **530**), for instance, by a supply wire that extends along the length of the catheter **1500**. The respective electrodes **1502**, **1504**, and **1506** of each of shock wave emitters **1501**, **1502**, and **1503** may be spaced apart by a respective spark gap **1510**, **1512**, and **1514** from an electrode **1508** that is connected to a negative terminal (e.g., return terminal) of the voltage source. Accordingly, electrode **1508** may form one of the electrodes at each respective shock wave emitter. The respective positive terminals connected to each of electrodes **1502**, **1504**, and **1506** may be simultaneously pulsed, resulting in a voltage applied across the respective spark gaps **1510**, **1512**, and **1514** separating each of electrodes **1502**, **1504**, and **1506** from electrode **1508**. The electrodes may be positioned such that when a sufficiently high voltage is applied across the electrodes **1502** and **1508**, **1504** and **1508**, and **1506** and **1508**, each shock wave emitter **1501**, **1503**, and **1505** generates a shock wave that propagates in a direction forward of the distal end of the catheter body. For instance, electrode **1508** may be positioned at an equal distance from each of electrodes **1502**, **1504**, and **1506**. Electrode **1508** may be positioned at the center of the distal end of catheter **1500**, and electrodes **1502**, **1504**, and **1506** may be positioned at equal distances from electrode **1508** about the circumference of the distal end of the catheter **1500**, as shown. Additionally, the shock wave emitters are configured such that when each simultaneously generates a shock wave, a single combined cavitation bubble may be formed rather than three separate cavitation bubbles that constructively interfere distally of the distal end of catheter **1500**. The voltage polarity (i.e., direction of current flow) between the respective electrodes of shock wave emitters **1501**, **1503**, and **1505** may be switched between voltage pulses. Such polarity switching may promote more uniform wear of electrodes and extend device longevity.

Although the electrode assemblies and catheter devices described herein have been discussed primarily in the context of treating coronary occlusions, such as lesions in vasculature, the electrode assemblies and catheters herein can be used for a variety of occlusions, such as occlusions in the peripheral vasculature (e.g., above-the-knee, below-the-knee, iliac, carotid, etc.). For further examples, similar designs may be used for treating soft tissues, such as cancer and tumors (i.e., non-thermal ablation methods), blood clots, fibroids, cysts, organs, scar and fibrotic tissue removal, or other tissue destruction and removal. Electrode assembly and catheter designs could also be used for neurostimulation treatments, targeted drug delivery, treatments of tumors in body lumens (e.g., tumors in blood vessels, the esophagus, intestines, stomach, or vagina), wound treatment, non-surgical removal and destruction of tissue, or used in place of thermal treatments or cauterization for venous insufficiency and fallopian ligation (i.e., for permanent female contraception).

In one or more examples, the electrode assemblies and catheters described herein could also be used for tissue engineering methods, for instance, for mechanical tissue decellularization to create a bioactive scaffold in which new cells (e.g., exogenous or endogenous cells) can replace the old cells; introducing porosity to a site to improve cellular retention, cellular infiltration/migration, and diffusion of nutrients and signaling molecules to promote angiogenesis, cellular proliferation, and tissue regeneration similar to cell replacement therapy. Such tissue engineering methods may be useful for treating ischemic heart disease, fibrotic liver, fibrotic bowel, and traumatic spinal cord injury (SCI). For instance, for the treatment of spinal cord injury, the devices and assemblies described herein could facilitate the removal of scarred spinal cord tissue, which acts like a barrier for neuronal reconnection, before the injection of an anti-inflammatory hydrogel loaded with lentivirus to genetically engineer the spinal cord neurons to regenerate.

The elements and features of the exemplary electrode assemblies and catheters discussed above may be rearranged, recombined, and modified, without departing from the present invention. Furthermore, numerical designators such as "first", "second", "third", "fourth", etc. are merely descriptive and do not indicate a relative order, location, or identity of elements or features described by the designators. For instance, a "first" shock wave may be immediately succeeded by a "third" shock wave, which is then succeeded by a "second" shock wave. As another example, a "third" emitter may be used to generate a "first" shock wave and vice versa. Accordingly, numerical designators of various elements and features are not intended to limit the disclosure and may be modified and interchanged without departing from the subject invention.

It should be noted that the elements and features of the example catheters illustrated throughout this specification and drawings may be rearranged, recombined, and modified without departing from the present invention. For instance, while this specification and drawings describe and illustrate catheters having several example balloon designs, the present disclosure is intended to include catheters having a variety of balloon configurations. The number, placement, and spacing of the electrode pairs of the shock wave generators can be modified without departing from the subject invention. Further, the number, placement, and spacing of balloons of catheters can be modified without departing from the subject invention.

It should be understood that the foregoing is only illustrative of the principles of the invention, and that various modifications, alterations and combinations can be made by those skilled in the art without departing from the scope and spirit of the invention. Any of the variations of the various catheters disclosed herein can include features described by any other catheters or combination of catheters herein. Furthermore, any of the methods can be used with any of the catheters disclosed. Accordingly, it is not intended that the invention be limited, except as by the appended claims.

While this disclosure has been particularly shown and described with references to embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the scope of the claims. For all of the embodiments described above, the steps of the methods need not be performed sequentially.

The invention claimed is:

1. A catheter for use in a body lumen, the catheter comprising:

a catheter body; and

a plurality of shock wave emitters disposed at a distal end of the catheter body, each shock wave emitter configured to generate a shock wave that propagates distally of the catheter body, wherein at least one shock wave emitter of the plurality of shock wave emitters comprises electrodes separated by a spark gap and at least one electrical connection to an electrode of at least one other shock wave emitter of the plurality of shock wave emitters, wherein an outermost surface of a shock wave emitter of the plurality of shock wave emitters is positioned externally relative an outer circumferential surface of the catheter body.

2. The catheter of claim **1**, wherein the plurality of shock wave emitters are arranged such that shock waves emitted from the plurality of shock wave emitters can constructively interfere distally of the catheter body.

3. The catheter of claim **1**, wherein the plurality of shock wave emitters are electrically connected in series such that an electrical pulse applied across an electrode of a first shock wave emitter of the plurality of shock wave emitters and an electrode at a second shock wave emitter of the plurality of shock wave emitters causes each of the plurality of shock wave emitters to emit a respective shock wave.

4. The catheter of claim **1**, wherein at least one shock wave emitter of the plurality of shock wave emitters can be driven independently of at least a second shock wave emitter of the plurality of shock wave emitters.

5. The catheter of claim **1**, wherein a first shock wave emitter of the plurality of shock wave emitters comprises an exposed distal tip of a first insulated wire and an exposed distal tip of a second insulated wire, the second insulated wire extending to a second shock wave emitter of the plurality of shock wave emitters.

6. The catheter of claim **5**, wherein the first insulated wire extends along the length of the catheter body and is positioned within a lumen of the catheter body.

7. The catheter of claim **1**, wherein a first shock wave emitter of the plurality of shock wave emitters comprises an exposed distal end of an insulated wire and a conductive emitter band that is separated from the exposed distal end of the insulated wire by a spark gap.

8. The catheter of claim **1**, wherein the plurality of shock wave emitters comprises at least three shock wave emitters.

9. The catheter of claim **1**, wherein the plurality of shock wave emitters are arrayed symmetrically about a longitudinal axis of the catheter body.

10. The catheter of claim **1**, wherein a distal most surface of a shock wave emitter of the plurality of shock wave emitters is flush with a distal most surface of the distal end of the catheter body.

11. The catheter of claim **1**, wherein a distal most surface of a shock wave emitter of the plurality of shock wave emitters is recessed from a distal most surface of the distal end of the catheter body.

12. The catheter of claim **1**, wherein a distal most surface of a shock wave emitter of the plurality of shock wave emitters is positioned forward of a distal most surface of the distal end of the catheter body.

13. The catheter of claim **1**, wherein the plurality of shock wave emitters are disposed at the same distal location relative to the distal end of the catheter body.

14. The catheter of claim **1**, further comprising an enclosure positioned to cover the plurality of shock wave emitters at the distal end of the catheter body.

15. The catheter of claim **1**, further comprising a central lumen extending from the proximal end of the catheter to the distal end of the catheter.

16. The catheter of claim **15**, wherein the central lumen is configured to receive a guide wire or a pacemaker wire lead.

17. The catheter of claim **1**, wherein the catheter body comprises an aspiration lumen.

18. The catheter of claim **1**, wherein the plurality of shock wave emitters are respectively spaced apart from one another by a distance of between 1 mm and 10 mm.

19. A method for removing a pacemaker lead comprising:

advancing a catheter along the pacemaker lead to a target site comprising fibrotic tissue, the catheter comprising a catheter body and a plurality of shock wave emitters disposed at a distal end of the catheter body, each shock wave emitter configured to generate a shock wave that propagates distally of the catheter body, wherein at least one shock wave emitter of the plurality of shock wave emitters comprises electrodes separated by a spark gap and at least one electrical connection to an electrode of at least one other shock wave emitter of the plurality of shock wave emitters; and

generating one or more shock waves to at least partially break up the fibrotic tissue so that the pacemaker lead can be removed.

20. The method of claim **19**, wherein the target site is within the heart.

21. The method of claim **19**, wherein the fibrotic tissue is located distally of the distal end of the catheter.

22. The method of claim **19**, wherein the pacemaker lead is inserted into a lumen of the catheter to guide the catheter to the target site.

23. The method of claim **19**, further comprising: advancing the catheter further along the pacemaker lead to the target site; and generating one or more additional shock waves.

24. The method of claim **19**, wherein the catheter further comprises a plurality of shock wave emitters positioned proximally of the distal end of the catheter body, wherein the second plurality of shock wave emitters are respectively configured to generate shock waves that propagate radially from the catheter body; and wherein the method further comprises:

generating a plurality of shock waves that propagate radially of the catheter using the second plurality of shock wave emitters to at least partially break up a calcified region of vasculature leading to the target site.

25. The method of claim **19**, wherein the plurality of shock wave emitters are electrically connected to a shock wave energy generator configured to deliver high voltage pulses to at least one shock wave emitter of the plurality of shock wave emitters.

26. The method of claim **25**, wherein the high voltage pulses are between 3 k V and 20 kV, including 3 kV and 20 kV.

27. The method of claim **25**, wherein the shock wave energy generator is configured to deliver the voltage pulses at a rate of up to 20 Hz, including 20 Hz.

28. The method of claim **25**, wherein the shock wave energy generator applies an alternating current to the electrodes to induce a change in the polarity of the electrodes.

29. The method of claim **19**, wherein the plurality of shock wave emitters are arranged such that shock waves emitted from the plurality of shock wave emitters can constructively interfere distally of the catheter body.

30. The method of claim **19**, wherein at least one shock wave emitter of the plurality of shock wave emitters can be driven independently of at least a second shock wave emitter of the plurality of shock wave emitters.

31. The method of claim **19**, wherein the plurality of shock wave emitters comprises at least three shock wave emitters.

32. The method of claim **19**, wherein the plurality of shock wave emitters are arrayed symmetrically about a longitudinal axis of the catheter body.

33. A catheter for use in a body lumen, the catheter comprising:

a catheter body; and

a plurality of shock wave emitters disposed at a distal end of the catheter body, each shock wave emitter configured to generate a shock wave that propagates distally of the catheter body, wherein at least one shock wave emitter of the plurality of shock wave emitters comprises electrodes separated by a spark gap, wherein a distal most surface of a shock wave emitter of the plurality of shock wave emitters is flush with a distal most surface of the distal end of the catheter body, and at least one electrical connection to an electrode of at least one other shock wave emitter of the plurality of shock wave emitters.

34. The catheter of claim **33**, wherein the plurality of shock wave emitters are arranged such that shock waves emitted from the plurality of shock wave emitters can constructively interfere distally of the catheter body.

35. The catheter of claim **33**, wherein the plurality of shock wave emitters are electrically connected in series such that an electrical pulse applied across an electrode of a first shock wave emitter of the plurality of shock wave emitters and an electrode at a second shock wave emitter of the plurality of shock wave emitters causes each of the plurality of shock wave emitters to emit a respective shock wave.

36. The catheter of claim **33**, wherein at least one shock wave emitter of the plurality of shock wave emitters can be driven independently of at least a second shock wave emitter of the plurality of shock wave emitters.

37. The catheter of claim **33**, wherein a first shock wave emitter of the plurality of shock wave emitters comprises an exposed distal tip of a first insulated wire and an exposed distal tip of a second insulated wire, the second insulated wire extending to a second shock wave emitter of the plurality of shock wave emitters.

38. The catheter of claim **33**, wherein a first shock wave emitter of the plurality of shock wave emitters comprises an exposed distal end of an insulated wire and a conductive emitter band that is separated from the exposed distal end of the insulated wire by a spark gap.

39. The catheter of claim **33**, wherein the plurality of shock wave emitters comprises at least three shock wave emitters.

40. The catheter of claim **33**, wherein the plurality of shock wave emitters are arrayed symmetrically about a longitudinal axis of the catheter body.

41. The catheter of claim **33**, wherein the plurality of shock wave emitters are disposed at the same distal location relative to the distal end of the catheter body.

42. The catheter of claim **33**, further comprising an enclosure positioned to cover the plurality of shock wave emitters at the distal end of the catheter body.

43. The catheter of claim **33**, further comprising a central lumen extending from the proximal end of the catheter to the distal end of the catheter.

44. The catheter of claim **43**, wherein the central lumen is configured to receive a guide wire or a pacemaker wire lead.

45. The catheter of claim **33**, wherein the catheter body comprises an aspiration lumen.

46. The catheter of claim **33**, wherein the plurality of shock wave emitters are respectively spaced apart from one another by a distance of between 1 mm and 10 mm.

47. A catheter for use in a body lumen, the catheter comprising:

a catheter body; and

a plurality of shock wave emitters disposed at a distal end of the catheter body, each shock wave emitter configured to generate a shock wave that propagates distally of the catheter body, wherein at least one shock wave emitter of the plurality of shock wave emitters comprises electrodes separated by a spark gap, wherein a distal most surface of a shock wave emitter of the plurality of shock wave emitters is positioned forward of a distal most surface of the distal end of the catheter body, and at least one electrical connection to an electrode of at least one other shock wave emitter of the plurality of shock wave emitters.

48. The catheter of claim **47**, wherein the plurality of shock wave emitters are arranged such that shock waves emitted from the plurality of shock wave emitters can constructively interfere distally of the catheter body.

49. The catheter of claim **47**, wherein the plurality of shock wave emitters are electrically connected in series such that an electrical pulse applied across an electrode of a first shock wave emitter of the plurality of shock wave emitters and an electrode at a second shock wave emitter of the plurality of shock wave emitters causes each of the plurality of shock wave emitters to emit a respective shock wave.

50. The catheter of claim **47**, wherein at least one shock wave emitter of the plurality of shock wave emitters can be driven independently of at least a second shock wave emitter of the plurality of shock wave emitters.

51. The catheter of claim **47**, wherein a first shock wave emitter of the plurality of shock wave emitters comprises an exposed distal tip of a first insulated wire and an exposed distal tip of a second insulated wire, the second insulated wire extending to a second shock wave emitter of the plurality of shock wave emitters.

52. The catheter of claim **51**, wherein the first insulated wire extends along the length of the catheter body and is positioned within a lumen of the catheter body.

53. The catheter of claim **47**, wherein a first shock wave emitter of the plurality of shock wave emitters comprises an exposed distal end of an insulated wire and a conductive emitter band that is separated from the exposed distal end of the insulated wire by a spark gap.

54. The catheter of claim **47**, wherein the plurality of shock wave emitters comprises at least three shock wave emitters.

55. The catheter of claim **47**, wherein the plurality of shock wave emitters are arrayed symmetrically about a longitudinal axis of the catheter body.

56. The catheter of claim **47**, wherein the plurality of shock wave emitters are disposed at the same distal location relative to the distal end of the catheter body.

57. The catheter of claim **47**, further comprising an enclosure positioned to cover the plurality of shock wave emitters at the distal end of the catheter body.

58. The catheter of claim **47**, further comprising a central lumen extending from the proximal end of the catheter to the distal end of the catheter.

59. The catheter of claim **58**, wherein the central lumen is configured to receive a guide wire or pacemaker lead.

60. The catheter of claim **47**, wherein the catheter body comprises an aspiration lumen.

61. The catheter of claim **47**, wherein the plurality of shock wave emitters are respectively spaced apart from one another by a distance of between 1 mm and 10 mm.

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